Introduction: Statistics and Economic Data

Statistics for Economists

Jisang Yu

LESE152

Course Logistics

Logistics at a Glance

- Instructor: Jisang Yu (`jisangyu@korea.ac.kr`) — office hours by appointment
- Lecture slides and materials posted on the course webpage before each class
- No required textbook. Free resources:
  - Krauth, *Introductory Statistics for Economics* (`bookdown.org/bkrauth/BOOK`)
  - Mahdi, *STAT 2601: Business Statistics* lecture notes
- Software: Microsoft Excel (university license) and R with RStudio (free) — bring a laptop to lab sessions (Weeks 2–3, 7, 12–13)

The syllabus is the authoritative document. Read it once, this week. Everything on the next two slides is a summary of what it already says.

Course Roadmap

1. **Describing data** (Weeks 1–3): descriptive statistics, Excel, R
2. **Probability** (Weeks 4–6): random events, random variables, distributions
3. **Inference** (Weeks 7–11): sampling, estimation, confidence intervals, hypothesis tests
4. **Regression** (Weeks 12–14): correlation, OLS, multiple regression, chi-square tests

Midterm in Week 8; Week 15 is make-up week; final exam in Week 16.

- The course is **cumulative**: each block builds directly on the one before it

Assessment and Deadlines

Weights

- Attendance 10%
- Assignments 40%
- Midterm 20%
- Final 20%
- Engagement 10%

Grades are **not curved**
(A+ $\geq$ 95, A $\geq$ 90, B+ $\geq$ 85, ...).

Four problem sets (each with an Excel/R component)

- PS1 Descriptive statistics & probability — Sep 25
- PS2 Random variables & sampling — Oct 16
- PS3 Inference & two-sample tests — Nov 13
- PS4 Regression analysis — Dec 4

Discussion is encouraged; write up and submit your **own** answers and code.

Attendance policy, academic integrity, and disability support: see the syllabus.

How to Succeed in This Course

- Statistics is learned by **doing**: work every example yourself, by hand and in software
- Do not fall behind — catching up in a cumulative course is expensive
- Form study groups, but write up problem sets independently
- Ask questions early; office hours by appointment
- Rough time budget: 3 hours in class + 4–5 hours outside per week

Now: why any of this is worth your time.

Why Statistics for Economists?

The Objective of the Course

- To build a working knowledge of **probability**, **statistical inference**, and **regression analysis** for analyzing economic data
- Statistics is the language of empirical economics:
  - How much does an additional year of education raise earnings?
  - Did a price increase reduce the quantity demanded, and by how much?
  - Is the observed difference between two groups real or just noise?
- We will combine **theory** (probability and inference) with **practice** (data analysis in Excel and R)

What is Statistics? What is Econometrics?

- **Statistics**: the science of collecting, describing, and interpreting data
- **Econometrics**: "the quantitative measurement and analysis of actual economic and business phenomena" (Studenmund)
- Major uses:
  1. **Description**: quantify relationships across economic variables
  2. **Inference**: test hypotheses about economic theories
  3. **Forecasting**: predict future economic variables
- This course builds the statistical foundation; econometrics courses build on it

Theory Needs Data, Data Need Theory

- Economic theory alone: tells us a demand curve slopes down, but not *how steeply*
- Data alone: patterns without interpretation — correlations that may mislead
- The scientific loop in economics:

$$\text{theory} \rightarrow \text{testable hypothesis} \rightarrow \text{data \& statistics} \rightarrow \text{revise theory}$$

- Example: Malthus predicted food per capita would fall; two centuries of yield data rejected the prediction — and the reasons (technology) reshaped growth theory

Example 1: Returns to Education

- Question: how much does one more year of schooling raise earnings?
- Naive answer: compare average wages of graduates vs. non-graduates
- Problem: graduates differ in *many* ways (family background, ability, location)
- Statistics gives us:
  - tools to **quantify** the raw gap (descriptive statistics)
  - tools to say how **precisely** it is measured (inference)
  - tools to **hold other factors constant** (multiple regression)

Example 2: Prices and Demand

- A retailer raises the price of coffee by 5% and sales fall by 2%
- Is the **price elasticity** of demand -0.4? Maybe — but:
  - Was there a simultaneous change in income, weather, or a competitor's price?
  - Would the same happen next month? Is -0.4 statistically distinguishable from 0?
- Later in the course you will estimate elasticities by regression and attach standard errors to them

Example 3: Evaluating a Policy

- A province introduces a fertilizer subsidy; average yields rise 8%
- Did the subsidy *cause* the increase? Alternative explanations:
  - Good weather that year
  - Farmers who applied were already more productive (**selection**)
- Modern economics answers with randomized experiments and careful observational designs — both built on the two-sample comparisons we develop later

Two Views of What Probability Means

- **Frequentist**: a probability is a **long-run relative frequency**
  - "$P(\text{heads}) = 0.5$" means: flip the coin forever and half the flips come up heads
  - The unknown parameter (μ, p, β) is a **fixed constant**; the **data** are random
  - Question asked: *what would happen if I repeated this study many times?*
- **Bayesian**: a probability is a **degree of belief**, updated by evidence
  - "$P(\text{rain tomorrow}) = 0.3$" — tomorrow happens once; there is no long run
  - The **data** are what they are; our **uncertainty about the parameter** is itself described by a probability distribution
  - Question asked: *what should I believe, given what I have seen?*

Both camps use the *same* rules of probability. They disagree about what the numbers are attached to.

Same Data, Two Answers

A new rice variety out-yields the old one on **7 of 10** test plots. Is it better?

	Frequentist	**Bayesian**
The unknown p is	a fixed constant we do not know	a quantity we are uncertain about
The question	How surprising is 7 of 10 *if* the two varieties were equal?	How likely is it that the new variety is better, *given* these plots?
The answer	p-value $= 0.17$: not unusual enough to reject "equal" at the 5% level	$P(\text{better} \mid \text{data}) = 0.89$, starting from a flat prior
The interval	95% **confidence** interval: 95% of intervals built this way would contain the true p	95% **credible** interval: given the data, p lies inside with probability 0.95

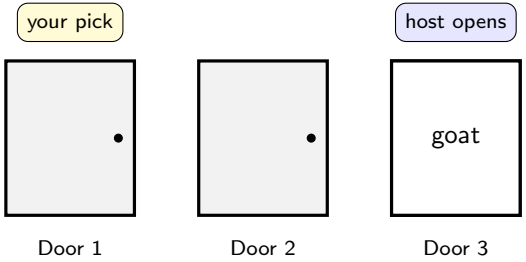
The two answers do not contradict each other — they answer **different questions**.

Where the Two Camps Came From

- **1763.** Thomas Bayes' essay is published two years after his death, by Richard Price. **Laplace** independently develops the same idea and spends 50 years applying it — to astronomy, to birth ratios, to court judgments.
- **19th century.** This *was* probability: it was called "inverse probability" and nobody thought it controversial. The objection was that the prior often had to be invented.
- **1920s–30s. R. A. Fisher** (likelihood, p-values, experimental design) and **Neyman & Pearson** (hypothesis tests, confidence intervals) build the frequentist framework *explicitly* to avoid priors. It wins the textbooks and the journals — partly on principle, largely because it could be computed with a table and a pencil.
- **1930s–50s.** Bayesian revival on philosophical grounds: **de Finetti**, **Jeffreys** (who argued the point with Fisher in print for years), and **Savage** show that coherent beliefs must obey the probability rules.
- **1990s–today.** Cheap computing and MCMC make Bayesian analysis practical; it is now standard in forecasting, macro (Bayesian VARs), and machine learning. Economics remains mostly frequentist — the tradition this course sits in.

A Puzzle: The Monty Hall Problem

A game show. Behind one of three doors is a car; behind the other two, goats.



1. You pick **Door 1**.
2. The host — who *knows* what is behind every door — opens **Door 3** and shows you a goat. He would never open the car.
3. He offers you a choice: **stick** with Door 1, or **switch** to Door 2.

Vote: stick, switch, or does it make no difference?

Monty Hall: Switch

Your first pick was Door 1. Three equally likely states of the world:

Car is behind	Probability	Host must open	Sticking wins?
Door 1	1/3	Door 2 or 3 (his choice)	✓
Door 2	1/3	Door 3 (forced)	—
Door 3	1/3	Door 2 (forced)	—

- **Sticking wins** 1/3 **of the time. Switching wins** 2/3**.** Switching doubles your chances.
- Your original door was right with probability 1/3, and *nothing the host does can change that* — he was always going to be able to show you a goat. So the remaining 2/3 collapses onto the one unopened door.
- The host's rule is what makes this work. If he opened a door *at random* and it happened to show a goat, it really would be 50–50.

Why Monty Hall Is on This Slide

- **Intuition about conditional probability is unreliable.** When Marilyn vos Savant published the correct answer in 1990, she received thousands of letters telling her she was wrong — many from people with PhDs. Paul Erdős, one of the great mathematicians of the century, reportedly stayed unconvinced until he was shown a simulation.
- **Evidence has to be conditioned on properly.** Formally this is Bayes' rule: $P(\text{car behind } 2 \mid \text{host opens } 3) = 2/3$. We will do this carefully when we take up conditional probability.
- **How the data were generated determines what you can conclude.** The numbers on the doors never changed — what mattered was the *rule* the host was following. Change the rule and the answer changes.
- That last point is the whole course in miniature: a dataset alone does not tell you what it means. You need to know where it came from.

Try It Yourself



`claude.ai/code/artifact/37f1b137-f427-49f2-a55e-0b2cd28e6afd`

If the scan opens the Claude app and stalls, type the address into a browser instead.

Monty Hall Lab — play the game and watch the tally build.

- Play ten rounds by hand. Ten is not many — notice how far from 2/3 you can land.
- Then let it run a thousand rounds and watch the two lines settle onto 1/3 and 2/3.

Data: The Raw Material

Populations and Samples

- **Population**: the entire set of units we want to learn about (all households, all firms, all transactions)
- **Sample**: the subset of the population we actually observe
- **Parameter**: a numerical feature of the population (usually unknown), e.g. μ, σ^2, p
- **Statistic**: a numerical feature of the sample, e.g. $\bar{x}$, s^2, $\hat{p}$
- Core problem of statistics: **what can we say about the population from a sample?**

Census vs. Sample

- Why not always observe the whole population?
  - Cost and time (a census of 20 million households is expensive)
  - Sometimes impossible (testing every battery destroys every battery)
  - Populations can be conceptual: all *potential* customers, all *future* harvests
- A well-drawn sample of 1,000 can beat a sloppy census: **how** you sample matters more than how much
- Korea's Population and Housing Census: conducted every 5 years, with the complete-enumeration component now register-based; quinquennial sample surveys provide additional detailed information. Most economic surveys are samples.

Descriptive vs. Inferential Statistics

- **Descriptive statistics**: summarizing and presenting the data at hand
  - Tables, graphs, means, standard deviations
- **Inferential statistics**: drawing conclusions about a population from a sample
  - Estimation, confidence intervals, hypothesis tests
- The bridge between them is **probability**: it tells us how samples behave when drawn from a known population — inference reverses the direction

Anatomy of a Dataset

Think of data as a rectangle: **rows** are units, **columns** are variables.

id	region	educ. level	n. children	income (10k won)	owns home
1	Seoul	high school	2	412.5	yes
2	Busan	bachelor's	0	587.0	no
3	Jeju	bachelor's	3	331.8	yes
⋮	⋮	⋮	⋮	⋮	⋮

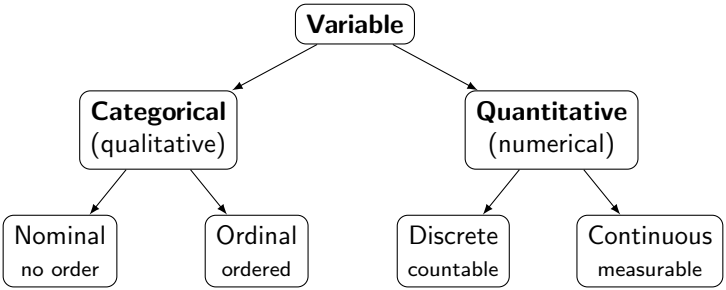
- **Unit of observation**: what one row *is* (here: a household) — always know this
- **Variable**: a characteristic that *varies* across units (a column)
- Notice: these five variables are five **different kinds** of thing. That distinction drives everything we do next.

Types of Economic Data: Structure

- **Cross-sectional data**: many units, one point in time
  - e.g., household expenditure survey for 2026
- **Time series data**: one unit, many points in time
  - e.g., monthly rice prices, 2000–2026
- **Panel (longitudinal) data**: many units, many points in time
  - e.g., annual output of 200 farms over 10 years
- This course focuses on cross-sectional tools; time-series and panel methods build on them

Types of Variables

The Big Picture



- First question to ask of *any* column of data: **which branch is this?**
- The answer determines what you may compute, what you may plot, and which test or model applies

Categorical Variables

Values are **labels**, not amounts. Arithmetic on them is meaningless.

- **Nominal** — categories with no natural order
  - region, industry (KSIC code), crop planted, occupation, marital status
  - only counts and proportions make sense: "38% of firms are in manufacturing"
- **Ordinal** — categories with a meaningful order, but *unknown or unequal spacing*
  - education level (HS $<$ BA $<$ MA $<$ PhD), credit rating (AAA $>$ AA $>$ A), Likert scales, income brackets
  - order is real: "at least a bachelor's degree" is a meaningful statement
- **Binary (dichotomous)** — a special case with exactly two categories
  - employed / not, owns home / not, treated / control
  - coded 0/1 it becomes a **dummy variable**, and then the mean *is* meaningful: it is the proportion

Quantitative Variables: Discrete vs. Continuous

Values are **amounts**; differences and (often) ratios are meaningful.

- **Discrete**: a countable set of values — usually the answer to "how many?"
  - number of children, employees, store visits, accidents per year
  - gaps between possible values are real: no household has 2.4 children
  - modeled with the discrete distributions we meet later (Bernoulli, binomial, Poisson)
- **Continuous**: any value in an interval — the answer to "how much?"
  - income, price, yield per hectare, time, temperature
  - any two values have a value between them; we only *record* it rounded
  - modeled with continuous densities (uniform, normal)

Borderline cases are normal. Money is technically discrete (won are whole units) but we treat it as continuous. A count with many possible values (population of a city) is treated as continuous too. The choice is about which *model* is convenient, not about metaphysics.

Measurement Scales: A Finer Cut

	Nominal	Ordinal	Interval	Ratio
Categories distinguishable ($=, \neq$)	yes	yes	yes	yes
Ordered ($<, >$)	no	yes	yes	yes
Differences meaningful ($-$)	no	no	yes	yes
True zero, ratios meaningful ($\div$)	no	no	no	yes
Mode	yes	yes	yes	yes
Median, percentiles	no	yes	yes	yes
Mean, standard deviation	no	(?)	yes	yes
Example	region	rating	°C, year	income, yield

- **Interval**: no natural zero — 20°C is not "twice as warm" as 10°C
- **Ratio**: natural zero — 40,000 won *is* twice 20,000 won
- Each scale permits everything the ones to its left permit, and more

Ordinal Data: Why the Mean is Slippery

Course satisfaction from 50 students. **Coding A** spaces the categories evenly; **Coding B** belongs to someone who thinks "good" → "excellent" is a bigger step than "poor" → "fair".

Response	very poor	poor	fair	good	excellent	**Mean**
Count	5	5	0	10	30	
Coding A	1	2	3	4	5	**4.10**
Coding B	0	1	2	4	8	**5.70**

- Only the arbitrary numbers changed, not the data — and neither coding is "wrong"
- The **median** and **mode** are "excellent" either way: they use only the *order*
- Economists often do average Likert scales — do it knowingly, and show the distribution too
- Safer: report percentages per category, or collapse to a binary ("good or better": 80%)

The Type Determines the Tool

Variable type	**Summarize with**	**Plot with**	**Later in the course**
Nominal	counts, proportions, mode	bar chart, pie	chi-square tests
Ordinal	median, percentiles, mode	bar chart	rank-based comparisons
Discrete	mean, sd, frequencies	bar chart, histogram	binomial, Poisson
Continuous	mean, sd, quantiles	histogram, box, scatter	normal, t, regression
Two variables at a time:			
Two continuous	correlation	scatterplot	regression
Two categorical	cross-tab	stacked bar	test of independence
Categorical $\times$ continuous	group means	side-by-side box	two-sample tests

This table is a preview of the entire semester — come back to it before the final.

Common Mistakes with Variable Types

- **Averaging category codes.** Region coded 1–17: the "mean region" of 8.3 means nothing. The single most common spreadsheet error.
- **Numbers that are not quantities.** Postal codes, student IDs, phone numbers, KSIC industry codes — numerals used as names.
- **Missing-value codes treated as data.** Surveys often code "refused" as -99 or 9999; average them in and your mean income is negative.
- **Throwing away information.** Binning a continuous variable (exact income $\rightarrow$ five brackets) is sometimes useful, but it is a one-way door: you cannot recover the detail.
- **Ignoring units and deflation.** "Price" in won 2005 and won 2026 are not the same variable.

Quick Check

Classify each variable: nominal, ordinal, discrete, or continuous?

1. A farm's crop: rice, barley, or soybean
2. Number of tractors owned
3. Harvest weight in kilograms
4. Firm size class: small / medium / large
5. Year a firm was founded
6. Unemployment rate of a province
7. Whether a household received a subsidy
8. A student's ID number
9. Bond credit rating (AAA, AA, ...)
10. Hours worked last week

Then: which of these could sensibly be averaged?

Where Data Come From

Experimental vs. Observational Data

→ 경제학 데이터 - 구조 기초 (확실한 시사점 제안

- **Experimental data**: the analyst controls assignment (clinical trials, A/B tests, field experiments) → 실험
  - Randomization makes treated and control groups comparable in expectation
- **Observational data**: the world assigns treatments (most economic data!)
  - Education, prices, policies are *chosen*, not randomly assigned → 관측
- Same formulas, very different interpretations — keep this distinction in mind all semester

Where Does Economic Data Come From?

- **Surveys and censuses**: household surveys, labor force surveys, agricultural censuses
- **Administrative records**: tax data, customs data, social insurance registers
- **Markets**: prices, quantities, financial transactions, scanner data
- **Experiments**: randomized controlled trials, lab experiments
- Public sources: KOSIS, Bank of Korea ECOS, World Bank, FAOSTAT, FRED, OECD.Stat
- Caution: **how** the data were collected matters for what we can conclude

How Statistics Can Mislead

- **Biased samples**: online polls, self-selected reviews
- **Misleading graphs**: truncated axes, distorted areas
- **Confusing correlation with causation**: ice cream sales and drowning deaths
- **Cherry-picking**: reporting only the favorable comparison or time window
- "There are three kinds of lies: lies, damned lies, and statistics" — our goal is to make you hard to fool

A Stylized Example: Simpson's Paradox

Inspired by the 1973 UC Berkeley graduate-admissions data; the numbers below are illustrative, not the original Berkeley data.

	Department A		Department B	
	Applied	Admitted	Applied	Admitted
Men	400	60%	100	25%
Women	100	65%	400	30%

- Women have the higher admission rate in **both** departments (65% vs. 60%, and 30% vs. 25%), yet the aggregate favors men:

$$\underbrace{\frac{240+25}{500} = 53\%}_{\text{men}} \qquad \text{vs.} \qquad \underbrace{\frac{65+120}{500} = 37\%}_{\text{women}}$$

- The reason: women applied disproportionately to Department B, which admits far fewer of its applicants
- Aggregation can reverse conclusions — conditioning and regression address this

Mathematical Toolkit

Summation Notation

- We write sums compactly:

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \cdots + x_n$$

- Rules we use constantly:
  - $\sum_i (ax_i + b) = a\sum_i x_i + nb$
  - $\sum_i (x_i + y_i) = \sum_i x_i + \sum_i y_i$
  - but $\sum_i x_i y_i \neq (\sum_i x_i)(\sum_i y_i)$ in general
- Exercise: show that $\sum_i (x_i - \bar{x}) = 0$ $\Rightarrow \sum_{i=1}^{n} x_i - n\bar{x}$

$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$ (simple mean (average))

$= \sum_{i=1}^{n} x_i - n \times \frac{\sum_{i=1}^{n} x_i}{n} = 0$

Functions, Logs, and Growth Rates

- We will use the natural log heavily:

$$\log(ab) = \log a + \log b, \qquad \log(a/b) = \log a - \log b$$

- Key approximation for small changes:

$$\log(x_1) - \log(x_0) \approx \frac{x_1 - x_0}{x_0} = \text{growth rate}$$

- Example: $\log(105) - \log(100) = 0.0488 \approx 5\%$
- This is why log-transformed regressions give percent interpretations

Sets and Notation Preview

- Sets: $A \cup B$ (union: or), $A \cap B$ (intersection: and), A^c (complement: not)
- These become the algebra of **events** when we build up probability
- Greek letters you will meet:
  - μ (population mean), σ (population sd), ρ (correlation)
  - β (regression coefficients), ε (error term), α (significance level)
- Convention: Greek/capital letters for population parameters and random variables, Latin/lowercase for sample statistics and realized data

$\mu, \sigma, \rho, \beta, \varepsilon, \alpha$

Doing Statistics Well

Anatomy of an Empirical Study

Every empirical paper — and your problem sets — follow the same skeleton:

1. **Question**: sharp, answerable with data
2. **Data**: source, sample, unit of observation, key variables *and their types*
3. **Descriptive statistics**: means, spreads, plots — know your data first
4. **Method**: comparison, test, or regression — with assumptions stated
5. **Results**: estimates *with* standard errors and intervals
6. **Interpretation**: magnitude, significance, caveats

Data Ethics and Privacy

- Microdata describe real people and firms: confidentiality is a legal and ethical duty
- Anonymization, aggregation, and secure data rooms are standard practice
- Report honestly: no selective deletion of observations, no undisclosed specification search
- Reproducibility: keep code and data so others can verify your results
- These norms are enforced — in academia, government, and industry alike

Where You Will Use This

- **Further courses**: econometrics, price analysis, development, finance, marketing research
- **Thesis / research**: any empirical project starts with this toolkit
- **Careers**: analyst roles in banks, consultancies, government ministries, cooperatives, agri-business, tech
- Employers consistently rank data literacy among the top skills for economics graduates
- R experience is a resume item in itself

In-Class Activity

1. Write down one economic question you care about
2. What would the ideal dataset look like? (unit of observation, variables, time span)
3. For each variable you listed, what **type** is it?
4. Is that data experimental or observational? What could bias a naive comparison?

We will revisit these questions when we reach two-sample comparisons and multiple regression.

Describing Economic Data: Descriptive Statistics and Visualization

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Tabular and Graphical Methods

Frequency Tables

- A **frequency table** lists each value (or bin) and how often it occurs
- **Relative frequency**: frequency divided by the number of observations n
- **Cumulative frequency**: running total — "how many observations at or below this value?"
- For continuous variables, group values into **bins** (classes) of equal width
- Rule of thumb: use roughly $\sqrt{n}$ bins (or $2^k \geq n$), then adjust for readability

Worked Example: Building a Frequency Table

Monthly incomes (10 households, in 10,000 KRW): 180, 220, 250, 260, 280, 310, 330, 380, 450, 940

Bin	Frequency	Relative freq.	Cumulative
100–300	5	0.50	0.50
300–500	4	0.40	0.90
500–700	0	0.00	0.90
700–900	0	0.00	0.90
900–1100	1	0.10	1.00

- The empty bins and the isolated top bin already reveal the **outlier** and the **right skew**

Graphs for One Variable

- **Bar chart**: frequencies of a categorical variable
- **Histogram**: binned frequencies of a continuous variable; shape shows *center*, *spread*, and *skewness*
- **Line graph**: values of a time series against time
- **Boxplot**: median, quartiles, and outliers at a glance
- **Pie charts**: generally avoid — humans compare lengths better than angles

Histogram Shapes

- **Symmetric**: mean $\approx$ median (heights, test scores)
- **Right-skewed** (long right tail): mean $>$ median (income, farm size, firm revenue)
- **Left-skewed** (long left tail): mean $<$ median (exam scores with a ceiling)
- **Bimodal**: two peaks — often a sign of **two mixed populations** (e.g., part-time and full-time workers' hours)
- Always plot before you summarize: one number cannot show two peaks

Honest and Dishonest Graphs

- **Truncated axes**: starting a bar chart at 90 instead of 0 makes a 2% difference look huge
- **Area distortions**: doubling both height and width of an icon quadruples its area
- **Cherry-picked windows**: a stock chart that starts at the trough
- **Missing baselines or labels**: if the axis is unlabeled, be suspicious
- Rules: label axes, show zero when levels matter, use consistent scales across panels

Measures of Central Tendency

Mean, Median, Mode

- **Sample mean**:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

- **Median**: middle value when data are sorted (50th percentile); for even n, average the two middle values
- **Mode**: most frequent value (the only option for nominal data)
- Symmetric data: mean = median = mode; skewed data: they diverge

Mean vs. Median: Robustness

$$\text{mean} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \text{median} = \frac{(n+1)}{2}$$

Incomes: 180, 220, 250, 260, 280, 310, 330, 380, 450, 940

- Mean $= 360$; median $= (280 + 310)/2 = 295$
- Replace 940 with 9,400: mean jumps to 1,206; median is **unchanged**
- The mean is sensitive to outliers; the median is **robust**
- That is why news reports cite *median* household income and *median* home prices
- But the mean has better mathematical properties for inference — we will use both

Weighted and Grouped Means

- **Weighted mean**: $\bar{x}_w = \dfrac{\sum_i w_i x_i}{\sum_i w_i}$
  - Your GPA; price indices (CPI weights goods by expenditure shares); survey weights
- **Grouped data**: with bin midpoints m_j and frequencies f_j,

$$\bar{x} \approx \frac{\sum_j f_j m_j}{n}$$

- Example: portfolio return $= \sum_j (\text{share}_j)(\text{return}_j)$ — a weighted mean

Percentiles and Quartiles

- The p-**th percentile**: the value below which p% of observations fall
- **Quartiles**: Q_1 (25th), Q_2 (median), Q_3 (75th)
- **Interquartile range**: $IQR = Q_3 - Q_1$, the spread of the middle half
- Widely used in economics: income deciles, top 1% share, P90/P10 wage inequality ratio
- Software note: there are several interpolation conventions; Excel's `PERCENTILE.INC` and R's default may differ slightly — do not panic

Measures of Dispersion

Why Dispersion Matters in Economics

- Two regions with the same *mean* income can differ hugely in *inequality*
- Two assets with the same *mean* return can differ hugely in *risk*
- Two suppliers with the same *mean* delivery time can differ in *reliability*
- Most economic decisions trade off the mean against the spread — we need to measure both

Variance and Standard Deviation

- **Sample variance**:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

- **Sample standard deviation**: $s = \sqrt{s^2}$, in the same units as the data
- Computational shortcut: $s^2 = \frac{1}{n-1}\left(\sum_i x_i^2 - n\bar{x}^2\right)$
- Why divide by $n-1$? Deviations from $\bar{x}$ (not μ) are systematically too small; $n-1$ corrects this so that s^2 is *unbiased* (Week 9)

Worked Example: Computing s by Hand

Yields (t/ha): 4, 5, 6, 7, 8 $\quad (n = 5,\ \bar{x} = 6)$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
4	-2	4
5	-1	1
6	0	0
7	1	1
8	2	4

$$s^2 = \frac{4 + 1 + 0 + 1 + 4}{5 - 1} = \frac{10}{4} = 2.5, \qquad s = 1.58$$

Comparing Variability: the Coefficient of Variation

- **Coefficient of variation**: $CV = s/\bar{x}$ — dispersion relative to the mean, unit-free
- Example: which price is more volatile?
  - Rice: $\bar{x} = 2{,}000$, $s = 200 \Rightarrow CV = 0.10$
  - Beef: $\bar{x} = 20{,}000$, $s = 1{,}200 \Rightarrow CV = 0.06$
- Beef has the larger s but rice is *relatively* more volatile
- Only meaningful for ratio-scale variables with $\bar{x} > 0$

z-Scores

- **z-score** of observation x_i:

$$z_i = \frac{x_i - \bar{x}}{s}$$

- Measures how many standard deviations an observation is from the mean — comparable across different variables
- Example: you scored 78 (class mean 70, sd 8) in micro and 85 (mean 80, sd 10) in stats
  - Micro: $z = 1.0$; stats: $z = 0.5 \Rightarrow$ relatively *better* in micro
- $|z| > 3$ is a common flag for outliers

Chebyshev's Inequality and the Empirical Rule

- **Chebyshev** (any distribution!): at least $1 - 1/k^2$ of observations lie within k sds of the mean
  - $k = 2$: at least 75%; $k = 3$: at least 89%
- **Empirical rule** (bell-shaped data): about 68% within $\pm 1s$, 95% within $\pm 2s$, 99.7% within $\pm 3s$

Skewness and Kurtosis

- **Skewness**: standardized third moment

$$\text{skew} = \frac{1}{n}\sum_i \left(\frac{x_i - \bar{x}}{s}\right)^3$$

  positive = long right tail, negative = long left tail
- **Kurtosis**: standardized fourth moment; heavy tails have kurtosis > 3 (the normal benchmark)
- Financial returns famously have negative skew and fat tails — crashes are more common than the normal predicts

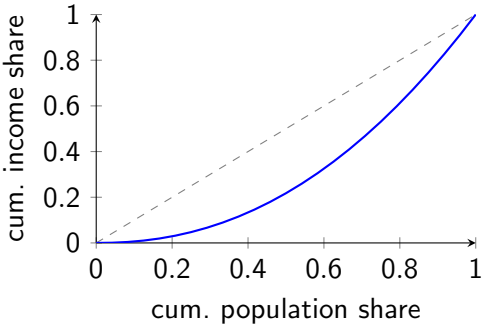
Boxplots and Outlier Detection

- The **boxplot**: box from Q_1 to Q_3, line at the median, whiskers to the last observation within $1.5 \times IQR$ of the box
- Points beyond the whiskers are flagged as **potential outliers**
- Income example: $Q_1 = 250$, $Q_3 = 380$, $IQR = 130$; upper fence $= 380 + 195 = 575 \Rightarrow$ 940 is flagged
- Outliers: first check for errors; if genuine, report results with and without them
- Side-by-side boxplots are excellent for comparing groups

Inequality Measures

The Lorenz Curve

- Sort households by income; plot cumulative population share (x) against cumulative income share (y)
- Perfect equality: the 45-degree line; more bowed = more unequal



The Gini Coefficient

- **Gini** = (area between the 45-degree line and the Lorenz curve) $\div$ (total area under the 45-degree line), between 0 (equality) and 1 (one person has everything)
- Ballpark values: Nordic countries ≈ 0.27, Korea ≈ 0.33, US ≈ 0.40, South Africa ≈ 0.63
- A single-number summary of an entire distribution — convenient, but different distributions can share a Gini
- Descriptive statistics for *distributions*, not just single variables, are central to economics

Two Variables

Covariance and Correlation

- **Sample covariance**:

$$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

- Sign shows direction of the linear association; magnitude depends on units
- **Sample correlation**:

$$r_{xy} = \frac{s_{xy}}{s_x s_y}, \qquad -1 \leq r_{xy} \leq 1$$

- Unit-free strength of *linear* association; unchanged by linear rescaling of either variable

Worked Example: Correlation by Hand

Price (x) and quantity (y) over 5 weeks:

x_i	10	11	12	13	14
y_i	100	95	88	82	75

- $\bar{x} = 12$, $\bar{y} = 88$; $\sum(x_i - \bar{x})(y_i - \bar{y}) = -62$
- $s_{xy} = -62/4 = -15.5$; $s_x = 1.58$, $s_y = 9.86$
- $r = -15.5/(1.58 \times 9.86) \approx -0.995$: an almost perfectly linear *negative* relationship — a demand curve in the wild

Interpreting Correlations

- Rough guide: $|r| < 0.3$ weak, 0.3–0.7 moderate, > 0.7 strong (context matters!)
- r only captures **linear** association: $y = x^2$ on symmetric data has $r = 0$
- r is distorted by outliers — always look at the scatter plot
- **Correlation is not causation**:
  - Third factors: ice cream and drownings (summer)
  - Reverse causality: police numbers and crime

Cross Tabulations and Conditional Averages

- **Cross tabulation (contingency table)**: joint frequencies of two categorical variables — tested formally in Week 14
- **Conditional averages**: mean of y within groups of x
  - Mean wage by education level; mean yield by seed variety
- In Excel: PivotTables; in R: `group_by() |> summarize()`

Lab: Descriptive Statistics in Excel

Data Cleaning Principles

- **Reproducibility**: keep the raw data untouched; document every change
- **Tidy data**: one row per observation, one column per variable
- Identify the **unit of observation** and a unique **identifier**
- Watch for: missing values, error codes (e.g., -999), text stored as numbers, inconsistent categories, duplicated rows

Key Excel Tools

- Navigation: sorting, filtering, freeze panes
- References: relative `A2` vs. absolute `$A$2`; fill-down
- Statistics: `AVERAGE`, `MEDIAN`, `STDEV.S`, `VAR.S`, `MIN`, `MAX`, `PERCENTILE.INC`, `COUNTIF`, `SUMPRODUCT` (weighted means), `CORREL`, `GEOMEAN`
- Charts: histogram, line, scatter; PivotTables for cross tabs and conditional means

In-Class Exercise

1. Clean: fix headers, remove duplicates, code missing values properly
2. Compute mean, median, sd, Q_1, Q_3, and CV of monthly earnings
3. Draw a histogram; describe the shape in one sentence
4. Flag outliers using the $1.5 \times IQR$ rule — are they errors or real?

Summary

- Describe one variable by **center** (mean, median, mode), **spread** (sd, IQR, CV), and **shape** (skewness, kurtosis)
- Use medians and IQRs for skewed data; geometric means for growth rates; weighted means for aggregates
- Describe two variables by **correlation**, **scatter plots**, and **conditional averages**

Describing Economic Data: Descriptive Statistics and Visualization

Statistics for Economists

Jisang Yu

LESE152

Tabular and Graphical Methods

Frequency Tables

- A **frequency table** lists each value (or bin) and how often it occurs
- **Relative frequency**: frequency divided by the number of observations n
- **Cumulative frequency**: running total — "how many observations at or below this value?"
- For continuous variables, group values into **bins** (classes) of equal width
- Rule of thumb: use roughly $\sqrt{n}$ bins (or $2^k \geq n$), then adjust for readability

Worked Example: Building a Frequency Table

Monthly incomes (10 households, in 10,000 KRW): 180, 220, 250, 260, 280, 310, 330, 380, 450, 940

Bin	Frequency	Relative freq.	Cumulative
100–300	5	0.50	0.50
300–500	4	0.40	0.90
500–700	0	0.00	0.90
700–900	0	0.00	0.90
900–1100	1	0.10	1.00

- The empty bins and the isolated top bin already reveal the **outlier** and the **right skew**

Graphs for One Variable

- **Bar chart**: frequencies of a categorical variable
- **Histogram**: binned frequencies of a continuous variable; shape shows *center*, *spread*, and *skewness*
- **Line graph**: values of a time series against time
- **Boxplot**: median, quartiles, and outliers at a glance
- **Pie charts**: generally avoid — humans compare lengths better than angles

Histogram Shapes

- **Symmetric**: mean $\approx$ median (heights, test scores)
- **Right-skewed** (long right tail): mean $>$ median (income, farm size, firm revenue)
- **Left-skewed** (long left tail): mean $<$ median (exam scores with a ceiling)
- **Bimodal**: two peaks — often a sign of **two mixed populations** (e.g., part-time and full-time workers' hours)
- Always plot before you summarize: one number cannot show two peaks

Honest and Dishonest Graphs

- **Truncated axes**: starting a bar chart at 90 instead of 0 makes a 2% difference look huge
- **Area distortions**: doubling both height and width of an icon quadruples its area
- **Cherry-picked windows**: a stock chart that starts at the trough
- **Missing baselines or labels**: if the axis is unlabeled, be suspicious
- Rules: label axes, show zero when levels matter, use consistent scales across panels

Measures of Central Tendency

Mean, Median, Mode

- **Sample mean**:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

- **Median**: middle value when data are sorted (50th percentile); for even n, average the two middle values
- **Mode**: most frequent value (the only option for nominal data)
- Symmetric data: mean = median = mode; skewed data: they diverge

Examples: Mean and Median

Mean

- Data 5, 7, 9, 12, 15: $\bar{x} = \dfrac{5+7+9+12+15}{5} = \dfrac{48}{5} = 9.6$
- Add an outlier, 5, 7, 9, 12, 15, 1000: $\bar{x} = \dfrac{1048}{6} \approx 174.67$ — sensitive to outliers

Median (M_d): sort the data, then find the position $\dfrac{n+1}{2}$

- n odd — data 5, 12, 15, 7, 9: sorted 5, 7, 9, 12, 15; position $6/2 = 3 \Rightarrow M_d = 9$
- n even — data 5, 7, 9, 12: position $5/2 = 2.5 \Rightarrow M_d = (7+9)/2 = 8$
- Resistant to outliers — data 1, 2, 3, 4, 5, 6, 100: $M_d = x_{(4)} = 4$

Examples: Median in Practice

Daily sales revenue (hundreds of \$) over 13 days: 6, 9, 8, 11, 10, 7, 12, 9, 8, 10, 13, 11, 9

- Sorted: 6, 7, 8, 8, 9, 9, **9**, 10, 10, 11, 11, 12, 13
- $n = 13$ is odd: median is the $(13+1)/2 = 7$th observation $\Rightarrow M_d = 9$

Customer waiting times (minutes): 8, 12, 15, 10, 18, 14, 20, 9, 16, 11, 13, 17

- Sorted: 8, 9, 10, 11, 12, **13**, **14**, 15, 16, 17, 18, 20
- $n = 12$: location $(12+1)/2 = 6.5 \Rightarrow$ average of the 6th and 7th observations

$$M_d = \frac{13+14}{2} = 13.5 \text{ minutes}$$

Examples: Mode and the Shape of a Distribution

The **mode** (M_o) is the most frequent value; useful for categorical or discrete data.

- 2, 3, 3, 4, 5, 6: $M_o = 3$ — **unimodal** 1개 최빈
- 1, 2, 2, 3, 4, 4, 5: modes = 2 and 4 — **bimodal** 2개 최빈
- 1, 1, 2, 2, 3, 3, 4: modes = 1, 2, and 3 — **multimodal** 3개 이상 최빈

Distribution	Visual shape	Relationship
Symmetric	Bell-shaped; mirror-image sides	$\bar{x} \approx M_d \approx M_o$
Right-skewed	Long tail to the right	$\bar{x} > M_d > M_o$
Left-skewed	Long tail to the left	$\bar{x} < M_d < M_o$

Mean vs. Median: Robustness

Incomes: 180, 220, 250, 260, 280, 310, 330, 380, 450, 940

- Mean $= 360$; median $= (280 + 310)/2 = 295$
- Replace 940 with 9,400: mean jumps to 1,206; median is **unchanged**
- The mean is sensitive to outliers; the median is **robust**
- That is why news reports cite *median* household income and *median* home prices
- But the mean has better mathematical properties for inference — we will use both

Weighted and Grouped Means

- **Weighted mean**: $\bar{x}_w = \dfrac{\sum_i w_i x_i}{\sum_i w_i}$
  - Your GPA; price indices (CPI weights goods by expenditure shares); survey weights
- **Grouped data**: with bin midpoints m_j and frequencies f_j,

$$\bar{x} \approx \frac{\sum_j f_j m_j}{n}$$

- Example: portfolio return $= \sum_j (\text{share}_j)(\text{return}_j)$ — a weighted mean

Percentiles and Quartiles

→ Excel data set 기능 사용

- The p-**th percentile**: the value below which p% of observations fall
- **Quartiles**: Q_1 (25th), Q_2 (median), Q_3 (75th)
- **Interquartile range**: $IQR = Q_3 - Q_1$, the spread of the middle half
- Widely used in economics: income deciles, top 1% share, P90/P10 wage inequality ratio
- Software note: there are several interpolation conventions; Excel's `PERCENTILE.INC` and R's default may differ slightly — do not panic

Percentiles by Linear Interpolation

Given sorted data $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$:

1. Position of the p-th percentile: $L_p = \dfrac{p}{100}(n+1)$
2. If L_p is an integer, the percentile is $x_{(L_p)}$; otherwise, with $k =$ integer part and $d =$ decimal part,

$$P_p = x_{(k)} + d\left(x_{(k+1)} - x_{(k)}\right)$$

Monthly car sales ($n = 10$): 12, 15, 18, 22, 25, 35, 40, 28, 30, 45

- Sorted: 12, 15, 18, 22, 25, 28, 30, 35, 40, 45
- $L_{30} = 0.30 \times 11 = 3.3$
- $P_{30} = 18 + 0.3(22 - 18) = 19.2$
- 30% of months had sales $\leq$ 19 cars

Commute times (min): 15, 20, 25, 30, 35, 40, 45, 50, 60, 70

- $L_{80} = 0.80 \times 11 = 8.8$ (between 50 and 60)
- $P_{80} = 50 + 0.8(60 - 50) = 58$
- 80% of employees commute $\leq$ 58 minutes; 20% commute longer

Example: Quartiles and the Five-Number Summary

Positions: $L_{Q_1} = \frac{n+1}{4}$, $L_{Q_2} = \frac{n+1}{2}$, $L_{Q_3} = \frac{3(n+1)}{4}$

Car sales (sorted): 12, 15, 18, 22, 25, 28, 30, 35, 40, 45

- $L_{25} = 0.25 \times 11 = 2.75$
  $Q_1 = 15 + 0.75(18 - 15) = 17.25$
- $L_{75} = 0.75 \times 11 = 8.25$
  $Q_3 = 35 + 0.25(40 - 35) = 36.25$

- Median $= (25 + 28)/2 = 26.5$
- $IQR = 36.25 - 17.25 = 19$

Minimum	Q_1	Median	Q_3	Maximum
12	17.25	26.5	36.25	45

- Five-number summary: center (median), spread of the middle 50% (IQR), and extremes (range)

Measures of Dispersion

Why Dispersion Matters in Economics

- Two regions with the same *mean* income can differ hugely in *inequality*
- Two assets with the same *mean* return can differ hugely in *risk*
- Two suppliers with the same *mean* delivery time can differ in *reliability*
- Most economic decisions trade off the mean against the spread — we need to measure both

Variance and Standard Deviation

- **Sample variance**:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

- **Sample standard deviation**: $s = \sqrt{s^2}$, in the same units as the data
- Why divide by $n-1$? Degrees of Freedom.

Example: Sample Variance — Two Formulas

Data: 5, 12, 6, 8, 14 $\quad (n = 5,\ \bar{x} = 45/5 = 9)$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
5	-4	16
12	3	9
6	-3	9
8	-1	1
14	5	25
45	0	60

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{60}{4} = 15$$

Worked Example: Computing s by Hand

Yields (t/ha): 4, 5, 6, 7, 8 $\quad (n = 5,\ \bar{x} = 6)$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
4	-2	4
5	-1	1
6	0	0
7	1	1
8	2	4

$$s^2 = \frac{4 + 1 + 0 + 1 + 4}{5 - 1} = \frac{10}{4} = 2.5, \qquad s = 1.58$$

Comparing Variability: the Coefficient of Variation

→ Scale

- **Coefficient of variation**: $CV = s/\bar{x}$ — dispersion relative to the mean, unit-free
- Example: which price is more volatile?
  - Rice: $\bar{x} = 2{,}000$, $s = 200 \Rightarrow CV = 0.10$
  - Beef: $\bar{x} = 20{,}000$, $s = 1{,}200 \Rightarrow CV = 0.06$
- Beef has the larger s but rice is *relatively* more volatile
- Only meaningful for ratio-scale variables with $\bar{x} > 0$

Example: Coefficient of Variation — Two Stocks

$$CV = \frac{s}{\bar{x}} \times 100\% \qquad \text{(lower CV = more stable)}$$

Stock A

- Average price last year = \$50
- Standard deviation = \$5
- $CV_A = \frac{5}{50} \times 100\% = 10\%$

Stock B

- Average price last year = \$100
- Standard deviation = \$5
- $CV_B = \frac{5}{100} \times 100\% = 5\%$

Both stocks have the same standard deviation, but stock B is less variable (more stable) relative to its price.

z-Scores

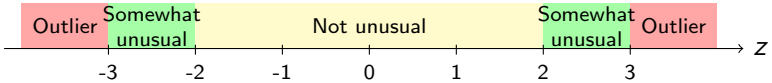
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  - Micro: $z = 1.0$; stats: $z = 0.5 \Rightarrow$ relatively *better* in micro

Example: z-Scores and Outlier Detection

- Population: $Z = \dfrac{X - \mu}{\sigma} \Rightarrow X = \mu + Z\sigma$; sample: $Z = \dfrac{X - \bar{x}}{s} \Rightarrow X = \bar{x} + Zs$
- Sign: above $(+)$ or below $(-)$ the mean; value: distance from the mean in standard deviations
- **IQ scores**, $\mu = 100$, $\sigma = 15$: for $X = 121$, $Z = (121 - 100)/15 = 1.4$ — 1.4 sds above the mean
- After conversion to z-scores, any data set has mean 0 and standard deviation 1



Chebyshev's Inequality and the Empirical Rule

- **Chebyshev** (any distribution!): at least $1 - 1/k^2$ of observations lie within k sds of the mean
  - $k = 2$: at least 75%; $k = 3$: at least 89%
- **Empirical rule** (bell-shaped data): about 68% within $\pm 1s$, 95% within $\pm 2s$, 99.7% within $\pm 3s$

Example: Empirical Rule

The time for a worker to complete an operation averages 12.8 minutes with a standard deviation of 1.7 minutes. The distribution is approximately mound-shaped. What proportion of workers take longer than 16.2 minutes?

- $\mu = 12.8$, $\sigma = 1.7$: $\quad 16.2 = \mu + 2\sigma = 12.8 + 2(1.7)$
- Empirical rule: 95% of observations lie within $\mu \pm 2\sigma$, i.e., between 9.4 and 16.2
- So 5% lie outside this interval; by symmetry, half of that is above 16.2

$$\text{Proportion} = \frac{0.05}{2} = 0.025$$

- About 2.5% of workers take longer than 16.2 minutes

Skewness and Kurtosis

- **Skewness**: standardized third moment

$$\text{skew} = \frac{1}{n}\sum_i \left(\frac{x_i - \bar{x}}{s}\right)^3$$

  positive = long right tail, negative = long left tail
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- Points beyond the whiskers are flagged as **potential outliers**
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- Outliers: first check for errors; if genuine, report results with and without them
- Side-by-side boxplots are excellent for comparing groups

Example: The $1.5 \times IQR$ Rule — Car Sales

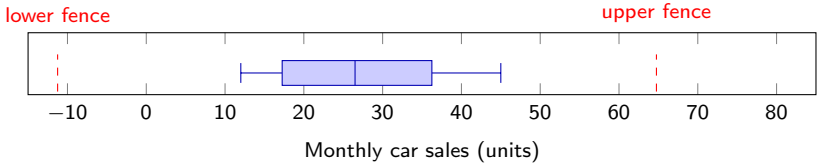
Fences: $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$; values outside are potential outliers

Car sales: 12, 15, 18, 22, 25, 28, 30, 35, 40, 45

- $Q_1 = 17.25$, $Q_3 = 36.25$, $IQR = 19$
- Fences: $17.25 - 28.5 = -11.25$ and $36.25 + 28.5 = 64.75$
- All data within the fences: **no outliers**

Replace 45 by 80: 12, 15, 18, 22, 25, 28, 30, 35, 40, 80

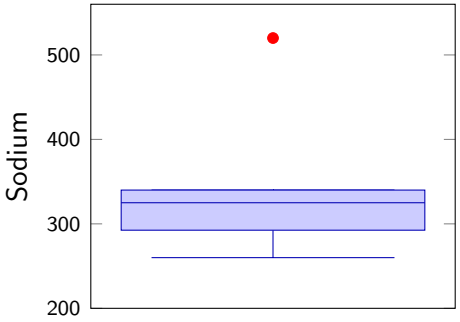
- Q_1, Q_3, IQR unchanged; $80 > 64.75$: **outlier detected**
- Whiskers: 12 to 40 (extremes inside the fences)



Example: Box Plot — Sodium in 8 Brands of Cheese

Sodium: 260, 290, 300, 320, 330, 340, 340, 520 $(n = 8)$

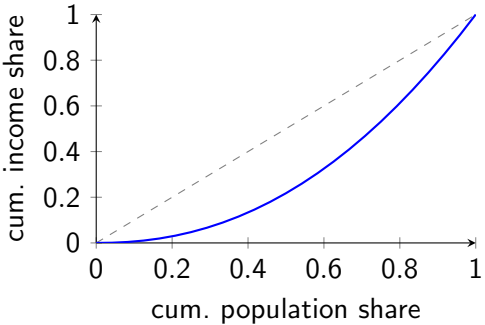
- $L_{Q_1} = 9/4 = 2.25$:
  $Q_1 = 290 + 0.25(300 - 290) = 292.5$
- $L_{Q_2} = 9/2 = 4.5$: $Q_2 = (320 + 330)/2 = 325$
- $L_{Q_3} = 27/4 = 6.75$:
  $Q_3 = 340 + 0.75(340 - 340) = 340$
- $IQR = 340 - 292.5 = 47.5$; $1.5 \times IQR = 71.25$
- Lower fence $= 292.5 - 71.25 = 221.25$: no outliers below
- Upper fence $= 340 + 71.25 = 411.25$: **520 is an outlier**
- Distribution is right-skewed



Inequality Measures

The Lorenz Curve

- Sort households by income; plot cumulative population share (x) against cumulative income share (y)
- Perfect equality: the 45-degree line; more bowed = more unequal



The Gini Coefficient

- **Gini** = (area between the 45-degree line and the Lorenz curve) $\div$ (total area under the 45-degree line), between 0 (equality) and 1 (one person has everything)
- Ballpark values: Nordic countries ≈ 0.27, Korea ≈ 0.33, US ≈ 0.40, South Africa ≈ 0.63
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$$r_{xy} = \frac{s_{xy}}{s_x s_y}, \qquad -1 \leq r_{xy} \leq 1$$

- Unit-free strength of *linear* association; unchanged by linear rescaling of either variable

Worked Example: Correlation by Hand

Price (x) and quantity (y) over 5 weeks:

x_i	10	11	12	13	14
y_i	100	95	88	82	75

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i-\bar{x})(y_i-\bar{y})$
10	100	10 - 12	12	
11	95	11 - 12	7	
12	88	0	0	$s_{xy} = \frac{-62}{4}$
13	82	13 - 12	-6	
14	75	14 - 12	-12	

- $\bar{x} = 12$, $\bar{y} = 88$; $\sum(x_i - \bar{x})(y_i - \bar{y}) = -62$
- $s_{xy} = -62/4 = -15.5$; $s_x = 1.58$, $s_y = 9.86$
- $r = -15.5/(1.58 \times 9.86) \approx -0.995$: an almost perfectly linear *negative* relationship — a demand curve in the wild

Interpreting Correlations

- Rough guide: $|r| < 0.3$ weak, 0.3–0.7 moderate, > 0.7 strong (context matters!)
- r only captures **linear** association: $y = x^2$ on symmetric data has $r = 0$
- r is distorted by outliers — always look at the scatter plot
- **Correlation is not causation**:
  - Third factors: ice cream and drownings (summer)
  - Reverse causality: police numbers and crime

Cross Tabulations and Conditional Averages

- **Cross tabulation (contingency table)**: joint frequencies of two categorical variables — tested formally in Week 14
- **Conditional averages**: mean of y within groups of x
  - Mean wage by education level; mean yield by seed variety
- In Excel: PivotTables; in R: `group_by() |> summarize()`

Lab: Descriptive Statistics in Excel

Data Cleaning Principles

- **Reproducibility**: keep the raw data untouched; document every change
- **Tidy data**: one row per observation, one column per variable
- Identify the **unit of observation** and a unique **identifier**
- Watch for: missing values, error codes (e.g., -999), text stored as numbers, inconsistent categories, duplicated rows

Key Excel Tools

- Navigation: sorting, filtering, freeze panes
- References: relative `A2` vs. absolute `$A$2`; fill-down
- Statistics: `AVERAGE`, `MEDIAN`, `STDEV.S`, `VAR.S`, `MIN`, `MAX`, `PERCENTILE.INC`, `COUNTIF`, `SUMPRODUCT` (weighted means), `CORREL`, `GEOMEAN`
- Charts: histogram, line, scatter; PivotTables for cross tabs and conditional means

In-Class Exercise

1. Find and download an economic datasets (try World Bank, KOSIS, FRED, FAOSTAT)
2. Compute mean, median, sd, Q_1, Q_3, and CV of the variable of interest
3. Draw a histogram; describe the shape in one sentence

Summary

- Describe one variable by **center** (mean, median, mode), **spread** (sd, IQR, CV), and **shape** (skewness, kurtosis)
- Use medians and IQRs for skewed data; geometric means for growth rates; weighted means for aggregates
- Describe two variables by **correlation**, **scatter plots**, and **conditional averages**

Introduction to R and Data Visualization

Statistics for Economists

Jisang Yu

LESE152

Where We Are Going Today

1. **Getting started** — install R, RStudio and the tidyverse; test that it works
2. **The language** — objects, types, vectors, factors, lists, missing values
3. **Wrangling** — the handful of `dplyr` verbs that do most of the work
4. **Visualization** — what makes a graph honest and readable, before any code
5. **ggplot2** — the grammar of graphics, and the traps: bins, overplotting, groups, layers
6. **A worked example** — is a warmer season good or bad for corn?
7. **Your turn** — corn-yield lab, then the in-class exercise

Today is hands-on. Open RStudio and type along — reading code is not learning code.

Getting Started: Installing R and RStudio

R and RStudio: Two Separate Programs

R — the engine

- The language that does the computing
- Download from CRAN: `https://cran.r-project.org`

RStudio — the dashboard

- The window you actually work in: editor, console, plots, files
- Download from Posit: `https://posit.co/download/rstudio-desktop/`

- Both are free, and run on Mac and Windows
- **Install R first, then RStudio.** RStudio finds R automatically
- After installing, you only ever open **RStudio** — it runs R for you

Step 1: Install R

Mac

1. Go to `https://cran.r-project.org` → *Download R for macOS*
2. Pick the installer that matches your chip (Apple menu → *About This Mac*):
   - Apple silicon (M1, M2, . . .): `R-4.x.x-arm64.pkg`
   - Intel: `R-4.x.x-x86_64.pkg`
3. Open the `.pkg` file and accept the defaults

Windows

1. Go to `https://cran.r-project.org` → *Download R for Windows* → *base*
2. Click *Download R-4.x.x for Windows*
3. Run the `.exe` file and accept the defaults

Install the **latest** release. If you already have R from another course, update it if its version is more than a year old — recent packages may refuse to install on an old R.

Step 2: Install RStudio Desktop

1. Go to `https://posit.co/download/rstudio-desktop/`
2. Step 1 on that page is R — you have already done it. Click the **RStudio Desktop** download button for your system
   - Mac: `RStudio-xxxx.dmg` — open it and drag RStudio into *Applications*
   - Windows: `RStudio-xxxx.exe` — run it and accept the defaults
3. Open **RStudio**. The **Console** should start with a line like
   `R version 4.x.x (20xx-xx-xx) -- "..."`

If RStudio says it cannot find R, R did not install correctly: reinstall R, then restart RStudio.

Step 3: Install the Packages We Use

Type this in the RStudio **Console** (bottom left) and press Enter:

```
install.packages(c("tidyverse", "readxl"))
```

- A lot of red and black text scrolls past — that is normal. Wait until the > prompt returns (one to several minutes)
- If asked *"Do you want to install from sources the packages which need compilation?"* type `no`
- If asked to choose a CRAN mirror, pick *0-Cloud* or *Korea*
- **Once per computer.** Every new session you only need `library(tidyverse)`

Step 4: Test Your Installation

Run each line in the Console and compare with the expected result:

```
1 + 1                              # [1] 2
R.version.string                   # "R version 4.x.x ..."
library(tidyverse)                 # a list of attached packages, no "Error"
packageVersion("tidyverse")        # '2.0.0' or newer

ggplot(data.frame(x = -5:5), aes(x = x, y = x^2)) +
  geom_line()                      # a parabola appears in the Plots pane
```

- `there is no package called 'tidyverse'` $\Rightarrow$ repeat Step 3
- All five work? You are ready for the rest of the course

No Installation? Posit Cloud

Posit Cloud (`https://posit.cloud`) runs RStudio in your web browser — nothing to install.

Getting started

1. Sign up for a free account
2. *New Project* → *New RStudio Project*
3. The same RStudio interface opens in your browser
4. **Files** pane: *Upload* data; *More* → *Export* to download results

Good to know

- Handy on a lab PC or when your own install breaks
- Run `install.packages()` again in each new project
- Free plan: monthly limits on usage time and memory
- Needs internet; files live on Posit's servers

Recommended: install R and RStudio on your laptop; keep Posit Cloud as the backup.

A Brief Tour of RStudio

- **Console** (bottom left): type commands, see results immediately
- **Source editor** (top left): write and save **scripts** (`.R` files)
- **Environment** (top right): objects currently in memory
- **Files / Plots / Packages / Help** (bottom right)
- Workflow: experiment in the console, save what works in a script
- Set up a **project** (File → New Project) so file paths stay simple

Run a line from the editor with `Cmd+Enter` (Mac) or `Ctrl+Enter` (Windows).

The R Language

Expressions and Assignment

```
2 + 3 * 4                          # 14

gdp <- 1900                        # assignment with <-
pop <- 52
gdp_per_capita <- gdp / pop
gdp_per_capita                     # 36.53846
```

- R is case-sensitive (`GDP` and gdp are different); # starts a comment
- Objects hold data; functions act on objects

Quick Check: Objects Are Labelled Boxes

```
x <- 3
y <- "abc"
y <- 5
x <- y
x + y       # ?
```

- A name on the outside, a value on the inside
- <- *copies* a value into the box; it does not link two boxes
- Reassigning replaces the old value, even with a different type
- Answer: `x + y` is 10

- Names start with a letter and use letters, digits, _ and .: `gdp_2024` works, `2024_gdp` does not
- The Environment pane shows every box; `rm(x)` removes one, `rm(list = ls())` removes all

Getting Help

```r
?mean                              # help page for a function
help.search("standard deviation")
example(mean)                      # run the examples
```

- Read the **Usage**, **Arguments**, and **Examples** sections of help pages
- Error messages: read them from the top; search the exact message
- AI assistants can draft code — but *you* must be able to verify it; exams are on paper

Vectors

```
price    <- c(2.5, 3.0, 2.8, 3.2)
quantity <- c(100, 80, 90, 70)
revenue  <- price * quantity      # elementwise

mean(price); sd(price); length(price)

price[2]              # indexing starts at 1
price[c(1, 3)]        # multiple elements
price[-1]             # everything except the first
price[price > 2.7]    # logical subsetting
years <- 2020:2026    # a sequence
```

- The vector is R's basic building block; most operations are **vectorized** — no loops needed
- Even a single number is a vector of length 1 — that is what the `[1]` in the output means

Vectors: Mistakes R Will Not Tell You About

```
odd <- seq(1, 99, by = 2)        # 50 elements
odd[0]                           # numeric(0) -- no error
odd[51]                          # NA         -- no error

c(1, 2, "a")                     # "1" "2" "a" -- everything became text

x <- -5:10                       # 16 elements
y <- seq(3, 30, by = 3)          # 10 elements
x + y                            # works, recycles y, only a warning
```

- Out-of-range indices and mixed types fail **silently**; unequal lengths are **recycled**
- This is why you inspect with `length()`, `class()`, `str()` after each step — R assumes you meant it

Data Types

```r
x <- c(1.5, 2, 3)                          # numeric
s <- c("Seoul", "Busan")                   # character
b <- c(TRUE, FALSE)                        # logical
f <- factor(c("low", "high", "low"))       # factor

class(x); str(f)
```

- **Factors** store categorical variables (with defined levels) — important for group comparisons and regression dummies later
- Logical vectors: `TRUE` counts as 1, so `mean(wage > 3000)` is the *share* above 3000 — a very useful idiom

123 vs. "123": Types and Coercion

```
typeof(123); typeof("123")     # "double"  "character"
"11" + "22"                    # Error: non-numeric argument
as.numeric("11") + as.numeric("22")   # 33

as.numeric("def")              # NA, with a warning
as.logical("123")              # NA -- no warning at all
as.numeric("005930")           # 5930 -- the leading zeros are gone
```

- Ask: *does arithmetic make sense?* Income, age, counts → numeric. Postal codes, student IDs, stock codes (Samsung is 005930), region codes → character
- `as.numeric()`, `as.character()`, `as.logical()` convert; failed conversions become `NA` — always check the result

Factors: When a Number Is Really a Category

```
cyl <- c(6, 4, 8, 4)
factor(cyl)                          # Levels: 4 6 8

day <- factor(c(1, 1, 2, 6),
              levels  = 1:7,
              labels  = c("Mon","Tue","Wed","Thu","Fri","Sat","Sun"),
              ordered = TRUE)
day                                  # Mon Mon Tue Sat;  Levels: Mon < Tue <
    ... < Sun
```

- **Continuous** variables live on the number line; **categorical** variables take a fixed set of values (*levels*)
- Codes stored as numbers (years, county FIPS codes, education level 1–5) must be converted with `factor()` or R treats them as continuous — we will *see* the damage in ggplot shortly
- `ordered = TRUE` for ordinal data (Week 1): plots and tables then follow the natural order

Missing Values

```
y <- c(4, NA, 6)

mean(y)                    # NA -- it infects everything
mean(y, na.rm = TRUE)      # 5 -- drop the missing value
is.na(y)                   # locate them
sum(is.na(y))              # count them
```

- `NA` = missing; most functions need `na.rm = TRUE`
- Always ask *why* data are missing — nonrandom missingness biases results (recall selection bias)

Lists, and What a Data Frame Really Is

```
student <- list(name    = "Kim",
                year    = 2,
                courses = c("ECON", "STAT", "MATH"))
student$courses[2]          # "STAT"
student[["name"]]           # "Kim"
length(student$courses)     # 3
```

- A **list** holds named elements of *any* type and length; a vector holds one type
- A **data frame** is a list of equal-length vectors: one vector per column — so `corn$value` is just a vector, and everything from the previous slides applies
- A **tibble** is the tidyverse data frame: prints compactly and shows each column's type (`<dbl>`, `<chr>`, `<fct>`)

Data Frames and Reading Data

```
library(tidyverse)
corn <- read_csv("corn_yield_weather_2013_2017.csv")

glimpse(corn)        # 6,945 rows, 11 columns: types and first values
head(corn)           # first rows
summary(corn)        # quick summary statistics
nrow(corn); ncol(corn)
```

- A **data frame**: rows = observations, columns = variables (tidy data!)
- Here one row is a **county-year**: US county corn yield plus that season's weather, 2013–2017
- After every import: check dimensions, types, and missing values — *before* anything else

Reading Other File Types

```
library(readxl)
d <- read_excel("data.xlsx")                          # Excel
d <- read_excel("data.xlsx", sheet = "Sheet2")        # a named sheet

d <- read_csv("data.csv")                             # comma separated
d <- read_delim("data.txt", delim = "\t")             # tab separated

library(haven)
d <- read_dta("data.dta")                             # Stata
```

- Path problems are the most common beginner error: use an RStudio **Project** and relative paths, never `setwd("/Users/yourname/...")`
- `getwd()` tells you where R thinks it is; `list.files()` shows what it can see

Packages and the Tidyverse

```
install.packages("tidyverse")    # once per computer
library(tidyverse)               # every session
```

- Packages extend R; CRAN hosts 20,000+
- The **tidyverse** bundles the main data tools: `dplyr` (wrangling), `ggplot2` (graphics), `readr` (import), `tidyr` (reshaping)
- Keep `library()` calls at the top of each script so dependencies are visible

Data Wrangling with dplyr

The Pipe and the Main Verbs

```
corn |>
  filter(year == 2017) |>                    # keep rows
  mutate(prec_in = prec / 25.4) |>           # new variable: mm -> inches
  select(state, county, value, prec_in) |> # keep columns
  arrange(desc(value))                       # sort: highest yield first
```

- The pipe |> passes the result to the next step: read the code as a recipe
- Five verbs cover 90% of data work: `filter`, `mutate`, `select`, `arrange`, `summarize`

Logical Conditions and Recoding

```
corn |> filter(state == "IOWA" & year >= 2015)
corn |> filter(state %in% c("IOWA", "ILLINOIS", "NEBRASKA"))

corn |>
  mutate(region = if_else(latitude >= 40, "North", "South"),
         heat   = case_when(hdd < 10  ~ "low",
                            hdd < 40  ~ "medium",
                            TRUE      ~ "high"))
```

- Comparison: ==, !=, >, >=; combine with & (and), | (or)
- `if_else()` creates a two-group variable; `case_when()` handles several cases, checked in order
- States are stored in capitals: `"Iowa"` matches nothing — look at the data first

Group Summaries

```
corn |>
  group_by(state) |>
  summarize(mean_yield = mean(value),
            sd_yield   = sd(value),
            n          = n()) |>
  arrange(desc(mean_yield))
```

- `group_by()` + `summarize()`: conditional statistics by group — Iowa and Illinois top the list (about 185 bu/acre); Texas is last (91)
- The R equivalent of an Excel PivotTable — but scriptable and reproducible
- Group by two variables for two-way tables: `group_by(state, year)`
- Give summaries **new names**: `summarize(hdd = mean(hdd), r = cor(hdd, value))` breaks, because the second line sees the *mean*, not the column

Joining Datasets

```
# yields from USDA, weather from a climate dataset -- two files
corn <- left_join(yields, weather, by = c("fips", "year"))
```

- Our corn file is already a merge: yields and weather come from different sources, matched on county (`fips`) and `year`
- `left_join()` keeps all rows of the first table and attaches matching columns from the second
- The `by` variables must identify units the same way in both tables: FIPS code 1001 vs. "01001" will not match
- Also: `inner_join` (intersection), `full_join` (union) — check row counts before and after

Reshaping: Long and Wide

```
wide <- corn |>
  select(fips, year, value) |>
  arrange(year) |>                  # columns in year order
  pivot_wider(names_from  = year, # one column per year
              values_from = value)

long <- wide |>                     # back to one row per county-year
  pivot_longer(cols = `2013`:`2017`,
               names_to = "year", values_to = "value")
```

- Spreadsheets are often **wide** (one column per year); analysis wants **long** (one row per unit-year)
- The wide table exposes gaps: 799 of 1,731 counties do not report all five years (`NA`)
- Tidy rule of thumb: if a variable (like year) is hiding in *column names*, reshape

Data Visualization Principles

Why Visualize Data?

A graph lets you:

- see patterns and trends far faster than in a table
- spot outliers, data-entry errors, and structural breaks
- explore relationships between variables
- communicate a finding to someone who will not read your regression table

"A picture is worth a thousand words" — but only if it is a *good* picture.

- A bad graph does not just fail to inform; it actively misleads
- You will plot your data *before* you compute anything. Today's example shows why.

Choosing the Right Chart

What you want to show	**Variables**	**Chart** (`geom`)
Distribution of one variable	continuous	histogram (`geom_histogram`)
Counts by category	categorical	bar chart (`geom_bar`)
Relationship between two variables	continuous vs. continuous	scatter plot (`geom_point`)
Change over time	continuous vs. time	line chart (`geom_line`)
Distributions across groups	continuous vs. categorical	box / violin (`geom_boxplot`)
Parts of a whole	categorical	bar chart (pie: sparingly)

The chart type follows from the **variable types** — exactly the taxonomy from Week 1.

Elements of a Good Graph

Every graph you submit needs all six:

1. **A clear title** — what is being shown?
2. **Labelled axes** — which variables?
3. **Units** — tonnes, won, percent, people
4. **An honest scale** — and a good reason for any truncation
5. **A legend**, if colour or shape carries meaning
6. **The source** of the data

Avoid: too many colours or patterns, decoration that encodes nothing, and 3-D effects on 2-D data.

Colour and Design

Use colour on purpose:

- Colour should encode something. If it encodes nothing, remove it.
- Three to five colours maximum; keep the same colour for the same category across all your figures
- Roughly 1 in 12 men has some colour-vision deficiency — avoid red/green contrasts, or use `scale_colour_viridis_d()`

Design:

- Remove everything that is not doing work (gridlines, borders, backgrounds)
- Text must be readable at the size it will actually be seen — projected, or in a printed paper
- Know your audience: a figure for your problem set is not a figure for a policy brief

Common Plotting Mistakes

1. **A truncated y axis with no good reason** — makes a small change look enormous
2. **Too many series in one panel** — if you need more than about five lines, facet instead
3. **Pie charts with many slices** — people cannot compare angles; use a bar chart
4. **Missing or cryptic labels** — your reader should never have to guess what a variable is
5. **Hiding the sample size** — a trend through six points is not the same as one through six hundred

These are the same honesty rules from Week 2, now under your own control.

Graphs with ggplot2

The Grammar of Graphics

Every ggplot is built from the same ingredients:

1. **Data** — a tidy table, one row per observation
2. **Aesthetic mappings** `aes()` — which column becomes x, y, colour, size, ...
3. **Geometry** `geom_*()` — points, lines, bars, boxes
4. **Scales and coordinates** `scale_*()`, `coord_*()` — log axes, limits, colour palettes
5. **Labels and theme** `labs()`, `theme_*()` — everything that is not data

```
ggplot(corn, aes(x = hdd, y = value)) +      # 1 + 2
  geom_point() +                             # 3
  scale_x_sqrt() +                           # 4
  labs(x = "Extreme heat degree days", y = "Yield (bu/acre)")    # 5
```

+ adds a layer. Change one ingredient and the rest stay the same.

The Basic Chart Types

```
ggplot(corn, aes(x = value)) + geom_histogram(bins = 30)      # one
    continuous
ggplot(corn, aes(x = state)) + geom_bar()                     # one
    categorical
ggplot(corn, aes(x = gdd, y = value)) + geom_point() +        # two
    continuous
  geom_smooth(method = "lm")                                  #   + fitted
    line
ggplot(iowa_by_year, aes(x = year, y = yield)) + geom_line()  # over time
ggplot(corn, aes(x = factor(year), y = value)) +              # continuous
  geom_boxplot()                                              #   by group
```

- Only the geom and the mapping change — the pattern is always `ggplot(data, aes(...)) + geom_*()`
- `geom_bar()` counts rows for you; use `geom_col()` when the heights are already in a column

Mapping vs. Setting: Inside or Outside aes()?

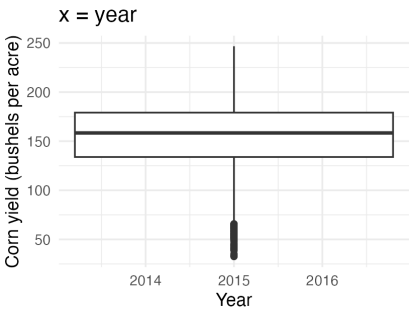
```
ggplot(corn, aes(x = gdd, y = value)) +
  geom_point(aes(color = state))    # MAPPING: from data -> legend

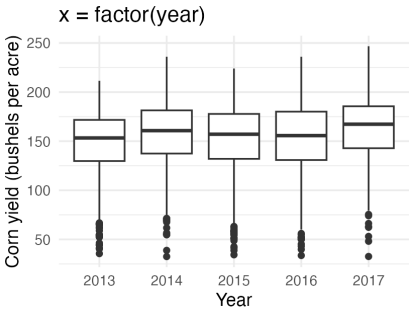
ggplot(corn, aes(x = gdd, y = value)) +
  geom_point(color = "navy")        # SETTING: constant -> no legend

ggplot(corn, aes(x = gdd, y = value)) +
  geom_point(color = state)         # Error: object 'state' not found
```

- Inside `aes()`: the value comes from a **column**. Outside: a **constant** you choose
- The last line fails because outside `aes()` R looks for an object called `state` in your environment, not in the data
- The same rule holds for `size`, `shape`, `alpha`, `linetype`, `fill`

A Number That Should Be a Category

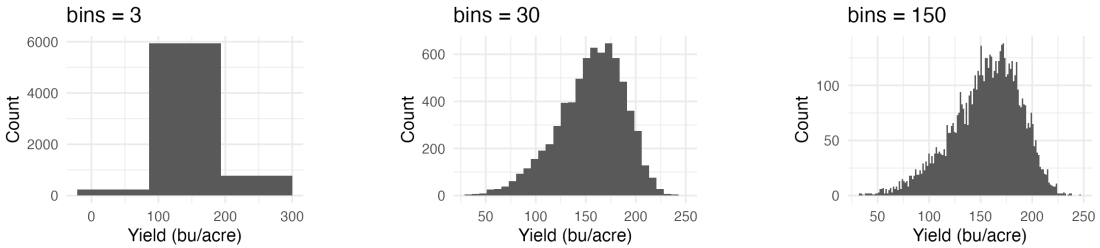




`year` is stored as a number, so `geom_boxplot()` sees one continuous x and draws a **single box** for all five years — with only a warning: `did you forget aes(group = ...)?` As a **factor**, each year gets its own box. The variable type decides the plot.

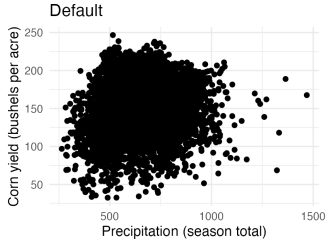
Histograms: The Number of Bins Is a Choice

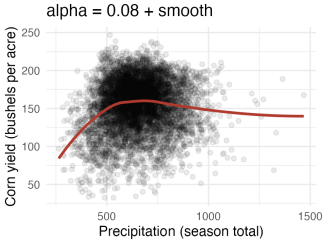
```
ggplot(corn, aes(x = value)) + geom_histogram(bins = 30)
```



- Too few bins hide the shape; too many turn it into noise. Try several; the default (30) is only a start
- At 30 bins the shape is clear: the middle half of county-years harvest 134–179 bu/acre, with a long **left tail** of failed or marginal harvests

Overplotting: 6,945 Points on One Plot

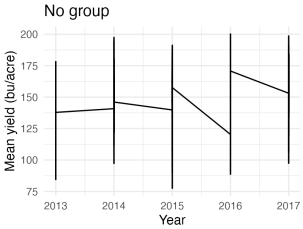


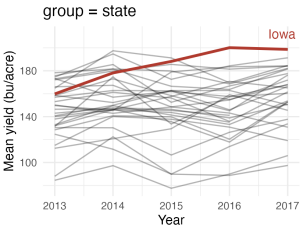



```
ggplot(corn, aes(x = prec, y = value)) +
  geom_point(alpha = 0.08) +                    # transparency
  geom_smooth(method = "loess", se = FALSE)     # the average pattern
```

On the left, a black blob. Transparency shows where the data are dense, and a smoother reveals the shape: yields rise with rain up to about 600–800 mm, then level off and fall — **too dry and too wet both hurt**. For a right-skewed axis, try `scale_x_log10()` or `scale_x_sqrt()`.

Line Plots Need to Know What to Connect



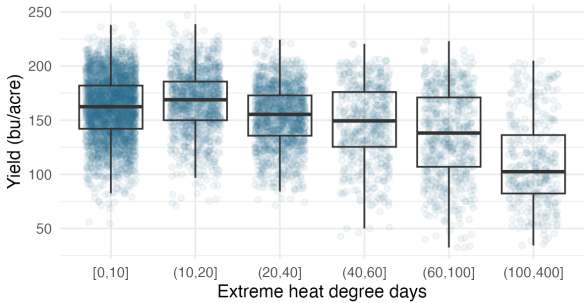



```
state_year <- corn |> group_by(state, year) |>
  summarize(yield = mean(value))
ggplot(state_year, aes(x = year, y = yield, group = state)) +
  geom_line(alpha = 0.3)
```

There are up to 30 states in each year. Without `group`, ggplot joins all of them into one zig-zag. With `group = state`, one line per state — and Iowa's county average climbs from 160 to about 200 bu/acre.

Layers: Summary on Top of the Raw Data

```
corn |>
  mutate(heat = cut(hdd,
    breaks = c(0, 10, 20, 40,
               60, 100, 400),
    include.lowest = TRUE)) |>
  ggplot(aes(x = heat,
             y = value)) +
  geom_jitter(width = 0.3,
              alpha = 0.06) +
  geom_boxplot(fill = NA,
    outlier.shape = NA)
```



Source: corn_yield_weather_2013_2017.csv

- Layers are drawn **in order**: later layers sit on top. `cut()` turns a number into bins (a factor)
- Mappings in `ggplot()` are **inherited** by every layer; put a mapping in one `geom` to apply it only there

Groups, Colours, and Facets

```
ggplot(state_year, aes(x = year, y = yield, color = state)) +
  geom_line()                                  # too many colours for 30
    states!

ggplot(corn, aes(x = value)) +
  geom_histogram() +
  facet_wrap(~ year)                           # one panel per year

ggplot(corn, aes(x = gdd, y = value)) +
  geom_point(alpha = 0.1) +
  facet_grid(region ~ year)                    # rows x columns

ggplot(corn, aes(x = region, fill = heat)) +
  geom_bar(position = "dodge")                 # side-by-side, not stacked
```

- Mapping a variable to `color` or faceting turns one plot into a comparison
- Faceting variables must be categorical; beyond about five colours, facet instead

Labels, Themes, and Saving

```
p <- ggplot(corn, aes(x = hdd, y = value)) +
  geom_point(alpha = 0.1) +
  labs(title    = "Extreme heat and corn yields, US counties, 2013-2017",
       x        = "Extreme heat degree days (growing season)",
       y        = "Corn yield (bushels per acre)",
       caption = "Source: corn_yield_weather_2013_2017.csv") +
  theme_minimal(base_size = 12)

ggsave("fig_heat.png", p, width = 7, height = 4.2, dpi = 200)
```

- `labs()` supplies five of the six required elements in one call
- Themes: `theme_minimal()`, `theme_bw()`, `theme_classic()` — pick one and stay with it
- `dpi = 300` for anything going into a printed document

A Worked Example: Weather and Corn Yields

The Question and the Data

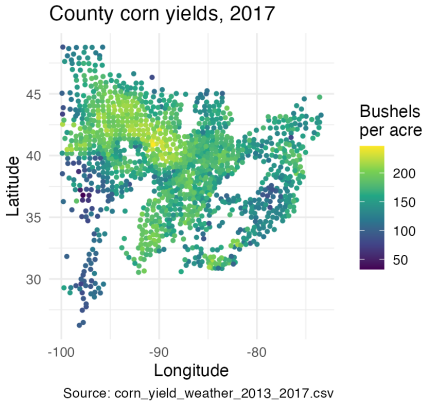
Is a warmer growing season good or bad for corn?

```
corn <- read_csv("corn_yield_weather_2013_2017.csv")
glimpse(corn)           # 6,945 rows, 11 columns
colSums(is.na(corn))    # no missing values
```

Variable	Meaning
`year`, `state`, `county`, `fips`	county-year identifiers (1,731 counties in 30 states)
`value`	corn yield, bushels per acre
`gdd`	growing degree days: accumulated warmth over the season
`hdd`	extreme heat degree days: accumulated heat above the damaging thresho
`prec`, `prec2`	season precipitation (mm) and its square
`latitude`, `longitude`	county location

Step 1: Where Is Corn Grown?

```
corn |>
  filter(year == 2017) |>
  ggplot(aes(x = longitude,
             y = latitude,
             color = value)) +
  geom_point(size = 1) +
  coord_quickmap() +
  labs(color = "Yield")
```

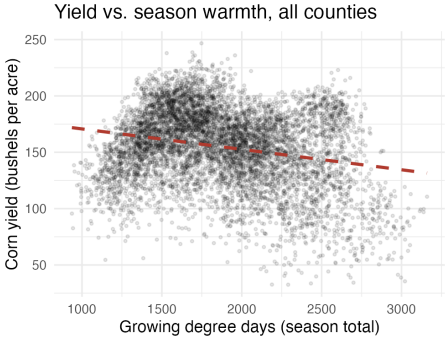


Latitude and longitude are enough for a map. The highest yields sit in the **Corn Belt** (Iowa, Illinois, Nebraska); the South and the Plains fringe harvest far less.

Step 2: Yield Against Season Warmth

```
ggplot(corn,
       aes(x = gdd, y = value)) +
  geom_point(alpha = 0.1) +
  geom_smooth(method = "lm",
              se = FALSE,
              linetype = "dashed")

cor(corn$gdd, corn$value)
#  -0.23
```



The correlation is -0.23. Read literally: *warmer seasons mean lower yields*. Should a farmer in Minnesota fear a warm summer?

Step 3: Split the Sample

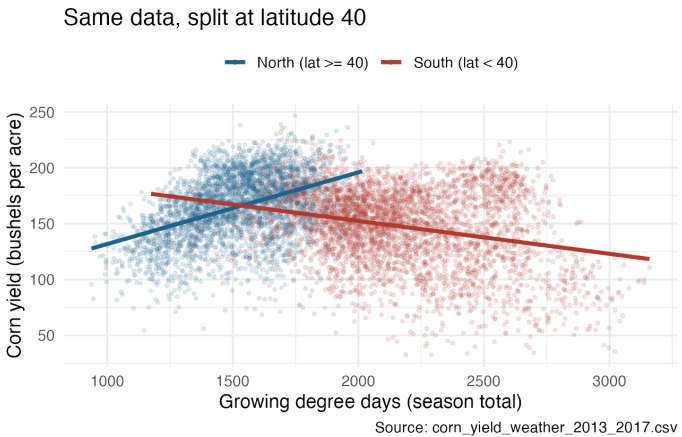
```
corn <- corn |>
  mutate(region = if_else(latitude >= 40, "North", "South"))

corn |> group_by(region) |> summarize(n = n(), r = cor(gdd, value))
```

Region	County-years	Mean yield	Mean GDD	Correlation
North (latitude ≥ 40)	3,118	164	1,507	$+0.42$
South (latitude < 40)	3,827	147	2,174	-0.27
All	**6,945**	**155**	**1,875**	$\mathbf{-0.23}$

Two relationships with *opposite signs*. In the cool North, extra warmth helps; in the warm South, it hurts. Pooled, the comparison is mostly *between* regions — the hot South also has lower yields — so the sign comes out negative.

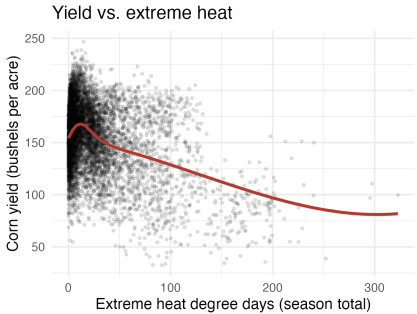
Step 3: What the Single Number Was Hiding



Step 4: The Real Damage Is Extreme Heat

```
ggplot(corn,
       aes(x = hdd, y = value)) +
  geom_point(alpha = 0.1) +
  geom_smooth(method = "loess",
              se = FALSE)

cor(corn$hdd, corn$value)
#  -0.40
```



Warmth is not the problem — **extreme heat** is. Yields hold up for the first 10–20 heat degree days and then fall steadily. This is the central finding of Schlenker & Roberts (2009, *PNAS*), and the reason economists worry about climate change and US agriculture.

Reproducible Workflow

Scripts, R Markdown, and Quarto

- A good script runs top-to-bottom in a **fresh session**: load packages → read data → clean → analyze → save output
- **R Markdown / Quarto**: text + code + results in one document that re-renders at the press of a button
- Problem sets may be submitted as rendered documents (HTML or PDF) — code and answers together
- Comment the *why*, not the obvious *what*

The one habit that matters most

Before you submit anything: `Session` → `Restart R`, then `Source`. If it does not run clean, it is not finished.

Common Beginner Errors (and Fixes)

- `could not find function` ⇒ package not loaded: `library(tidyverse)`
- `object not found` ⇒ typo, or the code that creates it was not run
- `unexpected symbol` ⇒ missing comma, unmatched parenthesis or quote — check the line *above* the one flagged
- File-not-found ⇒ wrong working directory: use RStudio projects
- Result is `NA` ⇒ missing values: add `na.rm = TRUE`
- `object 'x' not found` *inside a geom* ⇒ a column name used outside `aes()`
- A plot with a gradient legend for a category ⇒ convert with `factor()`
- Restarting R and re-running the script from the top fixes more than you would expect

Probability Functions Preview

- R has a family of functions for every distribution — we will use these from Week 5:
  - `d*`: density/PMF, e.g. `dbinom(8, 10, 0.7)`
  - `p*`: CDF, e.g. `pnorm(1.96)`
  - `q*`: quantiles, e.g. `qt(0.975, df = 24)`
  - `r*`: random draws, e.g. `rnorm(100)`
- Together with `replicate()`, the `r*` functions let us *simulate* sampling distributions — seeing the CLT instead of taking it on faith

Probability and Random Events

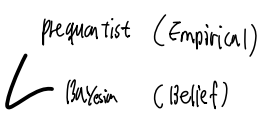
Jisang Yu

LESE152

Probability: Several Concepts

- Probability, Outcomes, and Events
- Rules of Probability
- Joint and Conditional Probability
- Independence
- Law of Total Probability and Bayes' Law
- Practice Questions

Probability



likelihood & Event
Random
→ deterministic

- Probability: A numerical measure of the likelihood that an event will occur. It ranges from 0 (impossible) to 1 (certain).
- Probability is a way of mathematically modeling a random process, so that we can understand it and make predictions about its future results.
- Casinos, banks, insurance companies, and any other businesses that manage risks rely on probability.
- Relationships between samples and populations are defined in terms of probability.

Assigning Probabilities

$P(A)$ = probability of A

$P(\text{dice} = 2) = \frac{1}{6}$

If) Rolling a dice: 1, 2, 3, 4, 5, 6

Sum of the numbers you get by rolling two different dice. $P(X=3) = \frac{2}{36}$

- Classical probability: If all outcomes are equally likely and the sample space is finite,

$$P(A) = \frac{\text{Number of outcomes favorable to } A}{\text{Total number of equally likely outcomes}}.$$

For example, $P(\text{Head}) = 1/2$ for a fair coin and $P(\text{Rolling a 5}) = 1/6$ for a fair die. What about $P(\text{Drawing an Ace from a deck of 52 cards})$? $4/52 = 1/13$.

- Empirical (relative frequency) probability: The observed frequency of A divided by the total number of trials. As the number of trials increases, the relative frequency stabilizes around the true probability (the Law of Large Numbers).
For example, if 47 out of 500 widgets are defective, $P(\text{Defective}) = 47/500 = 0.094$.

What Do Probabilities Mean?

- Frequentist interpretation: The random process can be repeated many times, and the probability of an event is the approximate fraction of times that it occurs. If you bet on Red in roulette 1000 times, you will win about 486 times.
- Bayesian interpretation: The random process is a one-time occurrence, and the probability represents the strength of our belief that the event will happen.
- The frequentist interpretation suits repeated settings like casino games or car insurance. The Bayesian interpretation makes more sense for predicting election results.

Roulette: An Example

- A roulette wheel has slots numbered 0 to 36. Slot 0 is green, 18 slots are red, and 18 slots are black.
- A ball is spun and lands in one slot. Players place bets such as:
  - Red (the ball lands in a red slot): pays \$1 per \$1 bet.
  - Black (the ball lands in a black slot): pays \$1 per \$1 bet.
  - A single number (the ball lands on that number): pays \$35 per \$1 bet.
- A roulette game is a random process. Something will happen, it matters to the players and the casino what will happen, but we do not know in advance what will happen.

Outcomes and the Sample Space

X = Number of 'Head's from 2 coin toss

Outcomes: 0, 1, 2

Simple space

- Outcome: A complete description of the result of a random process. Everything we are interested in should be defined in terms of the outcome.
- Sample space (Ω or S): The set of all possible outcomes. It must be exhaustive, and its outcomes must not overlap.
- Roulette: The outcome is the number of the slot in which the ball lands, b. The sample space is $\Omega = \{0, 1, 2, \ldots, 36\}$, which has $|\Omega| = 37$ elements.
- Other examples: $S = \{\text{Heads}, \text{Tails}\}$ for a coin flip; $S = \{\text{Defective}, \text{Non-Defective}\}$ for a product quality test; $S = \{2, 3, \ldots, 12\}$ for the total of two dice.

Events

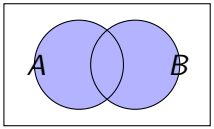
- Event (A): A subset of the sample space. Equivalently, a statement about the outcome that is either true or false.
- Simple (elementary) event: An event that contains only one outcome, e.g., rolling a 4 on a die, $\{4\}$.
- Compound event: A combination of two or more simple events, e.g., rolling an even number, $\{2, 4, 6\}$.
- Events in roulette:
  - The ball lands on 14: $b \in \{14\}$
  - The ball lands on red: $b \in Red = \{1, 3, 5, 7, 9, 12, 14, 16, 18, 19, 21, 23, 25, 27, 30, 32, 34, 36\}$
  - The ball lands on black: $b \in Black = \{2, 4, 6, 8, 10, 11, 13, 15, 17, 20, 22, 24, 26, 28, 29, 31, 33, 35\}$

Relationships Among Events

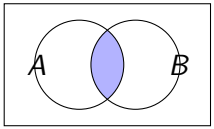
- Identical ($A = B$): The two events contain exactly the same outcomes. "The ball lands on 14" and "a bet on 14 wins" are identical.
- Implies ($A \subset B$): All outcomes of A are also in B. "The ball lands on 14" implies "the ball lands on red" since $\{14\} \subset Red$.
- Disjoint, or mutually exclusive ($A \cap B = \emptyset$): The two events share no outcomes, so they cannot both happen. $Red \cap Black = \emptyset$.
  But disjoint events can both fail to happen: if $b = 0$, neither red nor black wins.
- Any two elementary events are either identical or disjoint.

Venn Diagrams

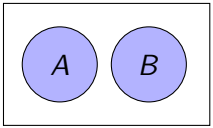
Union $A \cup B$



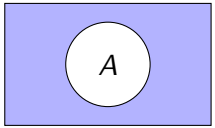
Intersection $A \cap B$



Mutually exclusive



Complement A^c

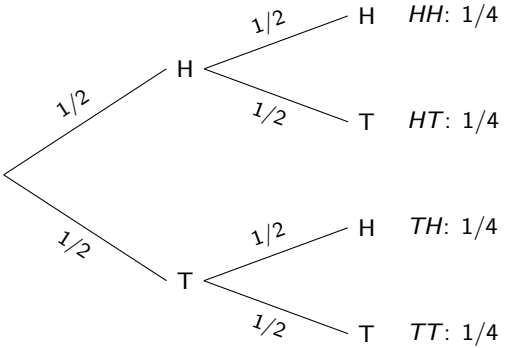


Throw two dice. Let A be "the first die is 1" and B be "the second die is 5."

- $A = \{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6)\}$ and $B = \{(1,5),(2,5),(3,5),(4,5),(5,5),(6,5)\}$.
- $A \cap B = \{(1,5)\}$, so $A \cup B$ has $6 + 6 - 1 = 11$ outcomes.

Roll one die. $A = \{1,3,5\}$ (odd) and $B = \{2,4,6\}$ (even) are mutually exclusive, and $A^c = B$.

Probability Tree: Two Coin Tosses



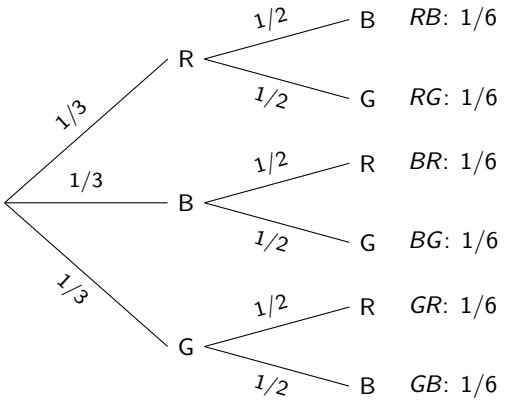
A probability tree represents the events of an experiment by lines. Toss a fair coin twice.

The probability of one H and one T: $P(HT) + P(TH) = 1/2$.

The probability of two T: $P(TT) = 1/4$.

What about the probability of at least one H?

Probability Tree: Selection Without Replacement



A bowl contains three M&Ms: one red (R), one blue (B), and one green (G). A child selects two at random without replacement.

The probability of no R:
$P(BG) + P(GB) = 1/3$.

What about the probability of at least one R?

The Axioms of Probability

- Probability distribution: A function that assigns a number, the probability, to each possible event.
- All valid probability distributions obey three axioms:
  1. Probabilities are never negative: $P(A) \geq 0$.
  2. One of the outcomes will definitely happen: $P(\Omega) = 1$.
  3. For any two disjoint events A and B: $P(A \cup B) = P(A) + P(B)$. $A \cup B$: A union B
- All other properties of probability can be derived from these three axioms.

Some Rules of Probability

Let A and B be two events. The three axioms imply:

- Probabilities cannot be higher than one: $0 \leq P(A) \leq 1$.
- The probability of nothing happening is zero: $P(\emptyset) = 0$.
- Complement rule: $P(A^c) = 1 - P(A)$.
- Identical events have the same probability, and if $A \subset B$ (belongs to), then $P(A) \leq P(B)$.
- Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
  If A and B are mutually exclusive, $P(A \cap B) = 0$ and $P(A \cup B) = P(A) + P(B)$.

Fair Roulette: An Example

Conditional = Joint / marginal

- Assume that the wheel is fair: each outcome has the same probability, p. This is an assumption, but a reasonable one because casinos are required by law to run fair wheels.
- Axiom 2 says $P(\Omega) = 1$. Since the outcomes are disjoint, axiom 3 says

$$P(\Omega) = P(\{0\}) + P(\{1\}) + \cdots + P(\{36\}) = 37p = 1.$$

  So $p = 1/37 \approx 0.027$.

- To find the probability of an event, add up the probabilities of its elementary events: $P(A) = |A| \times 1/37$.
- $P(Red) = 18/37 \approx 0.486$. What about $P(First12)$, where $First12 = \{1, 2, \ldots, 12\}$? $12/37 \approx 0.324$.

$|A|$ = number of outcomes lead to event A

Rules of Probability: Examples

- Roll a die. Let $A =$ "roll an even number" $= \{2, 4, 6\}$, so $P(A) = 3/6 = 1/2$. Then $P(A^c) = 1 - 1/2 = 1/2$.
- Roll a die. $P(\text{roll a } 10) = P(\emptyset) = 0$.
- An invoice is paid on time (A) with probability 0.65 and paid late (B) with probability 0.20. It cannot be both, so

$$P(A \cup B) = 0.65 + 0.20 = 0.85.$$

- Roll a die. Let $A = \{2, 4, 6\}$ and $B = \{3, 4, 5, 6\}$. Then $A \cap B = \{4, 6\}$ and

$$P(A \cup B) = 1/2 + 2/3 - 1/3 = 5/6.$$

Joint Probability

- We are often interested in how two events are related, e.g., a poker player after seeing the cards, or an economist predicting future economic conditions from current data.
- Joint probability: The probability that both A and B happen, $P(A \cap B)$.
- Roulette: What is the probability that the ball lands on a number that is both red and even?

$$P(Red \cap Even) = P(\{12, 14, 16, 18, 30, 32, 34, 36\}) = 8/37 \approx 0.216$$

- Joint probabilities are just probabilities, so they obey all the rules of probability.

Contingency Table: An Example

분할표

$P(\text{Man}) = \frac{227}{478}$ $P(\text{Ski}) = \frac{247}{478} = \frac{141}{478}, \frac{90}{478}$

Events can be placed in a contingency (two-way) table. Responses of 478 customers:

Gender	Skis	Camera	Bike	Total
Man	117	50	60	227
Woman	130	91	30	251
Total	247	141	90	478

각 범위 전체 부분

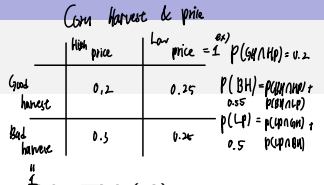
- Marginal probability (from the margins of the table): $P(\text{woman}) = 251/478 = 0.525$.
- Joint probability: $P(\text{woman} \cap \text{camera}) = 91/478 = 0.190$.
- Addition rule: $P(\text{woman} \cup \text{camera}) = (251 + 141 - 91)/478 = 0.63$.

MP written as Joint probability

$P(\text{woman}) = P(\text{woman} \cap \text{skis}) + P(\text{w} \cap \text{c}) + P(\text{w} \cap \text{B})$

$P(\text{Man} \cap \text{Ski}) = \frac{117}{478}, \frac{50}{478}, \frac{60}{478}$

Contingency Table: Cable Services



A cable company has 38 million cable passings. 10.9 million have only cable TV (A), 10.1 million have only cable internet (B), and 8.2 million have both.

	Internet, B	No internet, B^c	Total
Television, A	8.2	10.9	19.1
No television, A^c	10.1	8.8	18.9
Total	18.3	19.7	38

- Either service: $P(A \cup B) = (19.1 + 18.3 - 8.2)/38 \approx 0.77$.
- What about neither service? $P(A^c \cap B^c) = 8.8/38 = 0.23$.

Conditional Probability

- **Conditional probability:** The probability of A given that B has already occurred,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$

- It answers the question: if we already know that B is true, what are the chances that A is true?
- Examples: $P(\text{Recover from disease} \mid \text{Take the medicine})$, $P(\text{Hired} \mid \text{Pass the interview})$.
- Poker: At the start, every player has the same chance of winning. After you see your cards, your bet should be based on $P(\text{win} \mid \text{cards})$, not $P(\text{win})$.

Conditional Probability: Examples

	Remote work	In person	
w-f	0.25	0.45	0.7
w-p	0.15	0.15	0.3
	0.4	0.6	

$P(w\text{-}p \mid \text{in-person}) = \frac{0.15}{0.6}$

- Roulette:

$$P(Red \mid Even) = \frac{8/37}{18/37} \approx 0.444, \qquad P(b = 14 \mid Even) = \frac{1/37}{18/37} \approx 0.056$$

- Remote work: Let $A =$ "works remotely" and $B =$ "works full-time," with $P(A) = 0.40$, $P(B) = 0.70$, and $P(A \cap B) = 0.25$.

$$P(A \mid B) = \frac{0.25}{0.70} \approx 0.36$$

- Mutual funds: Let $A =$ "manager graduated from a top-20 MBA program" and $B =$ "fund outperforms the market," with $P(A \cap B) = 0.11$ and $P(A) = 0.40$.

$$P(B \mid A) = \frac{0.11}{0.40} = 0.275$$

Multiplication Rule

$P(A|B) = \frac{P(A \cap B)}{P(B)}$ / $P(A \cap B) = P(A|B) \cdot P(B)$ / $P(B \cap A) = P(B|A) \cdot P(A)$

- Rearranging the conditional probability gives the multiplication rule:

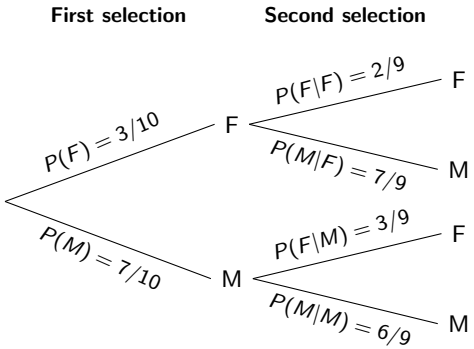
$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A).$$

- Some useful identities: (complement)
  - $P(A) = P(A \cap B) + P(A \cap B^c)$
  - $P(A \cup B) = 1 - P(A^c \cap B^c)$ and $P(A \cap B) = 1 - P(A^c \cup B^c)$
- In a company, 20% of employees work in IT (I), 30% hold a certification (C), and 50% of IT employees hold a certification. Then

$$P(I \cap C) = P(C \mid I)\, P(I) = 0.5 \times 0.2 = 0.1, \qquad P(I \mid C) = \frac{0.1}{0.3} = 0.33.$$

What about $P(I \cap C^c)$? $0.2 - 0.1 = 0.1$.

Probability Trees



- First branch: marginal probability. Second branch: conditional probability. Their product: joint probability.
- The branches from one node must add up to 1.
- A class has 7 male and 3 female students. Two are chosen without replacement. What is the probability that both are female?

$$P(F \cap F) = \frac{3}{10} \times \frac{2}{9} = \frac{6}{90}$$

Independent Events

- Independence ($A \perp B$): A and B are independent if

$$P(A \cap B) = P(A)\,P(B).$$

$P(A|B) = P(A)$ (ii) $P(A \cap B) = P(A)P(B)$

(i) $\frac{P(A \cap B)}{P(B)} = P(A)$

- If both events have nonzero probability, independence implies $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$. Knowing that one event is true tells you nothing about the other.
- Independence is reasonable when there is no physical or logical relationship between the events, usually because they are separated in time and space.
- Independence is not the same as mutually exclusive. Disjoint events have $P(A \cap B) = 0$.

Independence Across Roulette Games

Example Joint two Heads, four two coin toss

$P(H_1 \cap H_2) = P(H_1) \times P(H_2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

- Outcomes of two different roulette games can reasonably be assumed to be independent.
- You bring \$100 and bet everything on Red. If you win, you keep everything on Red for the next spin. After 3 games, you have \$800 if Red wins all 3 times, and \$0 otherwise.
- Because the games are independent,

$$P(w = \$800) = P(Red_1)\,P(Red_2)\,P(Red_3) = (18/37)^3 \approx 0.115.$$

- What about $P(w = \$0)$? $1 - 0.115 = 0.885$.

Checking Independence

- Events defined on the same outcome are typically related.
- Roulette: $P(Red) = 18/37 \approx 0.486$, but $P(Red \mid Even) = 8/18 \approx 0.444$. So Red and Even are not independent.
- Mutual funds: $P(B \mid A) = 0.275$, but $P(B) = 0.17$. So a top-20 MBA and outperforming the market are not independent.
- IT and certification: $P(I \mid C) = 0.33$, but $P(I) = 0.20$. Not independent.
- Independence is an assumption, and one that can easily be incorrect. Do not use results that only apply under independence when there is no reason to believe it holds.

Law of Total Probability

$$P(A) = \frac{P(A \cap B) \cdot P(B)}{P(B)} + \frac{P(A \cap B^c) P(B^c)}{P(B^c)}$$

$$= P(A \cap B) + P(A \cap B^c)$$

	H.P	L.P
GH	0.2	0.25
BH	0.3	0.25

$$P(GH \mid HP) = \frac{0.2}{0.5}$$

$$P(GH \mid HP^c) = \frac{0.25}{0.5}$$

- The law of total probability gives an unconditional probability from conditional probabilities:

$$P(A) = P(A \mid B)\, P(B) + P(A \mid B^c)\, P(B^c).$$

$$P(GH) = \frac{0.2}{0.5} \times 0.5 + \frac{0.25}{0.5} \times 0.5 = 0.45$$

- More generally, if $S_1, \ldots, S_k$ are mutually exclusive and exhaustive,

$$P(E) = P(E \mid S_1)P(S_1) + P(E \mid S_2)P(S_2) + \cdots + P(E \mid S_k)P(S_k).$$

- We create a set of scenarios, calculate the probability under each scenario, and add them up. This is useful for outcomes that occur in multiple stages.

Poker: An Example

- You are playing Texas hold'em, and one card is left to deal.
- If the last card is a heart (25% probability), you will have a flush and win with probability 90%. If not, you will win with probability 10%.
- Your overall chance of winning is

$$\begin{aligned} P(Win) &= P(Win \mid Hearts)\, P(Hearts) + P(Win \mid \text{not } Hearts)\, P(\text{not } Hearts) \\ &= 0.9 \times 0.25 + 0.1 \times 0.75 = 0.3. \end{aligned}$$

Bayes' Law

- Bayes' law restates one conditional probability in terms of the other:

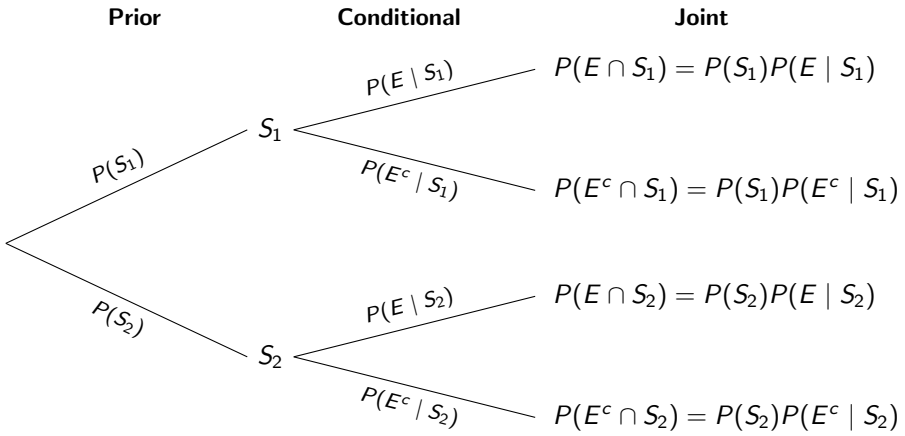
$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}.$$

- It is useful for evaluating evidence. Both Bayes' law and the law of total probability follow from the definition of conditional probability.
- More generally, with $S_1, \ldots, S_k$ mutually exclusive and exhaustive,

$$P(S_i \mid E) = \frac{P(E \mid S_i)\, P(S_i)}{\sum_{j=1}^{k} P(E \mid S_j)\, P(S_j)}.$$

$P(S_i)$ is the prior probability, $P(E \mid S_i)$ is the likelihood, and $P(S_i \mid E)$ is the posterior probability.

Bayes' Law: A Tree Diagram



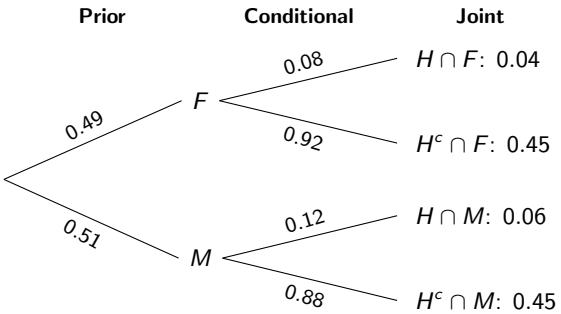
Medical Testing: An Example

- Let $T =$ "the patient tests positive" and $D =$ "the patient has the disease." No test is perfect.
- Sensitivity: $P(T \mid D) = p$. Specificity: $P(T^c \mid D^c) = q$. Prevalence: $P(D) = d$.
- A patient tests positive. What is $P(D \mid T)$? Bayes' law and the law of total probability give

$$P(D \mid T) = \frac{P(T \mid D)\,P(D)}{P(T)} = \frac{pd}{pd + (1-q)(1-d)}.$$

- Suppose $p = 1$ and $q = 0.99$. If the disease is common ($d = 0.10$), $P(D \mid T) \approx 0.917$. What about a rare disease ($d = 0.001$)? $P(D \mid T) \approx 0.091$.
- The same test means very different things depending on prevalence. A small false positive rate matters a lot when prevalence is low.

Heart Attack Risk: An Example



49% of the population are female. 8% of females and 12% of males are high risk for a heart attack (H).

A randomly selected person is high risk. What is the probability that this person is male?

$$P(M \mid H) = \frac{P(H \cap M)}{P(H \cap M) + P(H \cap F)}$$

$$= \frac{0.06}{0.06 + 0.04} = 0.6$$

Quality Control: An Example

- A company has three production lines. Line S_1 produces 70% of output with a 4% defect rate, S_2 produces 20% with a 2% defect rate, and S_3 produces 10% with a 96% defect rate.
- A randomly selected product is defective (H). What is the probability that it came from S_1?
- Joint probabilities: $P(H \cap S_1) = 0.028$, $P(H \cap S_2) = 0.004$, $P(H \cap S_3) = 0.096$.
- Law of total probability: $P(H) = 0.028 + 0.004 + 0.096 = 0.128$.
- Bayes' law: $P(S_1 \mid H) = 0.028/0.128 \approx 0.219$, down from the prior of 0.70.

Summary

- Probability rules: $0 \leq P(A) \leq 1$, $P(A^c) = 1 - P(A)$, and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Conditional probability: $P(A \mid B) = P(A \cap B)/P(B)$, and multiplication rule: $P(A \cap B) = P(A \mid B)\, P(B)$.
- Independence: $P(A \cap B) = P(A)\, P(B)$, equivalently $P(A \mid B) = P(A)$. Independence is not the same as mutually exclusive.
- Law of total probability: $P(E) = \sum_i P(E \mid S_i)\, P(S_i)$.
- Bayes' law: $P(S_i \mid E) = P(E \mid S_i)\, P(S_i)/P(E)$.

Practice Question 1: Dish Detergent

$P(A) = 0.65$ $P(B) = 0.48$ $P(A \cap B) = 0.25$

	A	A^c	
B	0.25	0.23	0.48
B^c	0.4	0.12	0.52
	0.65	0.35	

A company launched two new dish detergents. Among sampled customers, 65% liked detergent A, 48% liked detergent B, and 25% liked both.

(a) State the three probabilities in terms of the events A and B.

(b) Construct a 2×2 contingency table with rows A, A^c and columns B, B^c.

(c) What is the probability that a customer likes only detergent A?

(d) What is the probability that a customer likes at least one detergent?

(e) What is the probability that a customer likes neither detergent?

(f) What is the probability that a customer likes only one detergent?

(g) Are A and B mutually exclusive? NO. $P(A \cap B) = 0.25$

Practice Question 1: Answers

(a) $P(A) = 0.65$, $P(B) = 0.48$, $P(A \cap B) = 0.25$.

(b)

	B	B^c	Total
A	0.25	0.40	0.65
A^c	0.23	0.12	0.35
Total	0.48	0.52	1

(c) $P(A \cap B^c) = P(A) - P(A \cap B) = 0.40$.

(d) $P(A \cup B) = 0.65 + 0.48 - 0.25 = 0.88$.

(e) $P(A^c \cap B^c) = 1 - P(A \cup B) = 0.12$.

(f) $P(A \cap B^c) + P(A^c \cap B) = 0.40 + 0.23 = 0.63$.

(g) No, because $P(A \cap B) = 0.25 \neq 0$.

Practice Question 2: Credit Score and Loan Default

In a survey, 60% of people had a good credit score. Of those with a good credit score, 8% defaulted on a loan. Of those without a good credit score, 25% defaulted.

Let C = "has a good credit score" and D = "defaulted on a loan."

(a) Express the three probabilities in terms of C and D.

(b) What is the probability that a randomly selected person defaulted on a loan?

(c) If a person defaulted on a loan, what is the probability that the person did not have a good credit score?

Practice Question 2: Answers

(a) $P(C) = 0.60$, $P(D \mid C) = 0.08$, $P(D \mid C^c) = 0.25$.

(b) Law of total probability:

$$\begin{aligned} P(D) &= P(D \mid C)P(C) + P(D \mid C^c)P(C^c) \\ &= 0.08 \times 0.60 + 0.25 \times 0.40 = 0.048 + 0.10 = 0.148 \end{aligned}$$

(c) Bayes' law:

$$P(C^c \mid D) = \frac{P(C^c \cap D)}{P(D)} = \frac{0.10}{0.10 + 0.048} \approx 0.68$$

Practice Question 3: The Pass Bet

The main bet in craps is Pass. On the first ("come out") roll c, Pass wins if c is 7 or 11 and loses if c is 2, 3, or 12. Otherwise c becomes the "point," and the shooter keeps rolling: Pass wins if the point comes up again before a 7, and loses otherwise.

c	2	3	4	5	6	7	8	9	10	11	12
$P(c)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$
$P(PassWins \mid c)$	0	0	$\frac{3}{9}$	$\frac{4}{10}$	$\frac{5}{11}$	1	$\frac{5}{11}$	$\frac{4}{10}$	$\frac{3}{9}$	1	0

Use the law of total probability to calculate $P(PassWins)$.

Practice Question 3: Answer

$$
\begin{aligned}
P(PassWins) &= P(PassWins \mid c=2)P(c=2) + \cdots + P(PassWins \mid c=12)P(c=12) \\
&= \left(0 \times \tfrac{1}{36}\right) + \left(0 \times \tfrac{2}{36}\right) + \left(\tfrac{3}{9} \times \tfrac{3}{36}\right) + \left(\tfrac{4}{10} \times \tfrac{4}{36}\right) + \left(\tfrac{5}{11} \times \tfrac{5}{36}\right) + \left(1 \times \tfrac{6}{36}\right) \\
&\quad + \left(\tfrac{5}{11} \times \tfrac{5}{36}\right) + \left(\tfrac{4}{10} \times \tfrac{4}{36}\right) + \left(\tfrac{3}{9} \times \tfrac{3}{36}\right) + \left(1 \times \tfrac{2}{36}\right) + \left(0 \times \tfrac{1}{36}\right) \\
&= \frac{244}{495} \approx 0.4929
\end{aligned}
$$

- This is less than 50%, so the house still has an advantage.
- But it is higher than the probability of winning a bet on Red in roulette, $18/37 \approx 0.486$.

Practice Question 4: Returning Planes

During World War II, the statistician Abraham Wald analyzed bullet damage on returning planes. A plane can be hit on the engine ($EngineHit$) or on the body ($BodyHit$), and it either returns safely ($Return$) or crashes. Suppose that:

- $P(EngineHit) = P(BodyHit) = 0.6$
- Many of the returning planes show body damage: $P(BodyHit \mid Return) = 0.7$
- Few of the returning planes show engine damage: $P(EngineHit \mid Return) = 0.1$
- Most of the planes return: $P(Return) = 0.8$

Calculate $P(Return \mid BodyHit)$ and $P(Return \mid EngineHit)$.

Practice Question 4: Answer

Bayes' law:

$$P(Return \mid BodyHit) = \frac{P(BodyHit \mid Return)\,P(Return)}{P(BodyHit)} = \frac{0.7 \times 0.8}{0.6} \approx 0.93$$

$$P(Return \mid EngineHit) = \frac{P(EngineHit \mid Return)\,P(Return)}{P(EngineHit)} = \frac{0.1 \times 0.8}{0.6} \approx 0.13$$

- A plane hit on the body usually returns, but a plane hit on the engine usually does not.

Discrete Random Variables

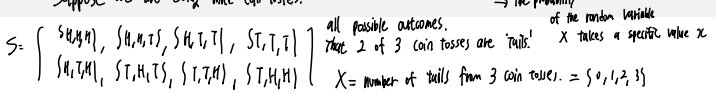
Jisang Yu

LESE152

Discrete Random Variables: Several Concepts

- Random Variables
- The PMF and the CDF
- Expected Value
- Variance and Standard Deviation
- Standard Discrete Distributions
- Practice Questions

Random Variables



- Random variable (X): A number whose value depends on a random outcome. Formally, it is a function that assigns a number to each outcome in the sample space.
- Economics is a mostly quantitative field: prices, quantities, interest rates, unemployment rates, and GDP are all numbers. When a random outcome is described by a number, we call it a random variable.
- Notation: A capital letter X denotes the random variable, and a lowercase letter x denotes a particular value it can take.
- Examples: The number of heads in 10 coin flips, a farm's harvest, next month's sales, or an indicator equal to 1 if a loan defaults and 0 otherwise.

Random Variables in Roulette

Recall the roulette wheel from Lecture 4: the outcome is the slot $b \in \{0, 1, \ldots, 36\}$. We can define several random variables from the same outcome.

- The number the ball lands on: b.
- An indicator for whether a bet on red wins:

$$r = \begin{cases} 1 & \text{if } b \in Red \\ 0 & \text{otherwise} \end{cases}$$

- The net profit from a \$1 bet on red, y_{Red}: $+1$ if the ball lands on red and -1 otherwise.
- The net profit from a \$1 bet on 14, y_{14}: $+35$ if the ball lands on 14 and -1 otherwise.

All four are functions of the same outcome b, so they are related to each other.

Discrete and Continuous Random Variables

- Support: The set of values that the random variable can actually take, i.e., the values with positive probability.
- Roulette: The support of b is $\{0, 1, \ldots, 36\}$, the support of r is $\{0, 1\}$, and the support of y_{14} is $\{-1, 35\}$.
- Discrete random variable: The possible values can be counted or listed, with gaps in between. Examples: the number of customers arriving in an hour, the number of defective items in a batch.
- Continuous random variable: The possible values fill an interval, so they cannot be listed. Examples: a price, a wage, the amount of rainfall. We study these in Lecture 6.

Probability Mass Function (PMF)

- Probability mass function: The probability that X takes each value,

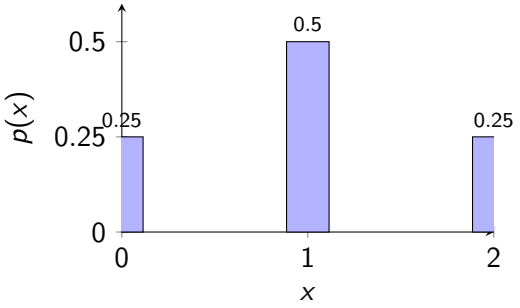
$$p(x) = P(X = x).$$

- It can be written as a table, a graph, or a formula. For a valid PMF:
  1. $0 \leq p(x) \leq 1$ for every x.
  2. $\sum_x p(x) = 1$, where the sum is over the support.
- The probability of any event is the sum of the PMF over the values in that event:

$$P(X \in A) = \sum_{x \in A} p(x).$$

This is the same rule as in Lecture 4: add up the probabilities of the elementary events.

PMF: Two Coin Tosses



Toss a fair coin twice. The sample space is $\{HH, HT, TH, TT\}$. Let $X =$ the number of heads.

x	0	1	2
$p(x)$	0.25	0.50	0.25

$p(1) = P(HT) + P(TH) = 0.5.$

What is $P(X \geq 1)$? $0.50 + 0.25 = 0.75.$

PMF: Roulette

Assume a fair wheel, so each slot has probability $1/37$.

- The PMF of b: $p_b(x) = 1/37$ for each $x \in \{0, 1, \ldots, 36\}$.
- The PMF of r:

$$p_r(1) = P(Red) = 18/37 \approx 0.486, \qquad p_r(0) = 19/37 \approx 0.514.$$

- The PMF of y_{14}:

$$p_{14}(35) = P(b = 14) = 1/37 \approx 0.027, \qquad p_{14}(-1) = 36/37 \approx 0.973.$$

- Mode: The most likely value. The mode of r is 0, and the mode of y_{14} is -1: the most likely result of either bet is losing.

Cumulative Distribution Function (CDF)

- Cumulative distribution function: The probability that X is at most x,

$$F(x) = P(X \leq x) = \sum_{s \leq x} p(s).$$

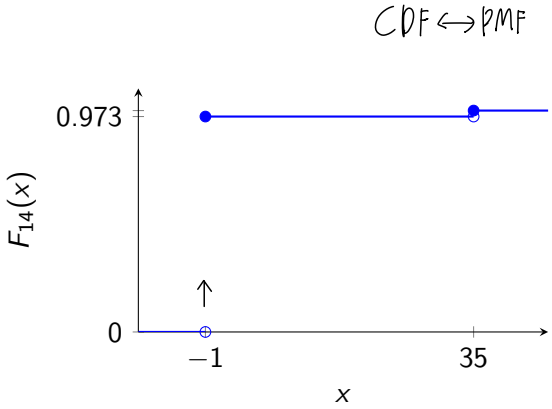
- For a discrete random variable, the CDF is a step function. It jumps up by $p(x)$ at each value in the support and is flat in between.
- Properties: $0 \leq F(x) \leq 1$, F never decreases, and F runs from 0 (far left) to 1 (far right).
- Two coin tosses:

	$250	$500	$1,000
$p(X=x)$	0.3	0.5	0.2
$F(X=x)$	0.3	0.8	1

$F(X=m)=1$

x	0	1	2
$p(x)$	0.25	0.50	0.25
$F(x)$	0.25	0.75	1

CDF: The Bet on 14



The support of y_{14} is $\{-1, 35\}$, so

$$F_{14}(x) = \begin{cases} 0 & x < -1 \\ 36/37 \approx 0.973 & -1 \le x < 35 \\ 1 & x \ge 35 \end{cases}$$

The CDF jumps at each point in the support, and the size of each jump equals the PMF at that point.

Interval Probabilities

$P(\$250 < X \le \$1{,}000) = F(\$1{,}000) - F(\$250)$

$\neq P(\$250 < X < \$1{,}000)$ $= 1 - 0.3 = 0.7$

$F(x=2) = P(X \le 2)$

- The CDF gives the probability of any interval:

$$P(a < X \le b) = F(b) - F(a).$$

- Be careful with $<$ and $\le$: a discrete random variable can equal the endpoint exactly. Adjust with the PMF, e.g.,

$$P(a \le X \le b) = F(b) - F(a) + p(a).$$

- Roulette: $F_b(1) = 2/37$ and $F_b(36) = 1$, so

$$P(1 < b \le 36) = 1 - 2/37 = 35/37, \qquad P(1 \le b \le 36) = 35/37 + 1/37 = 36/37.$$

- Complement rule: $P(X > a) = 1 - F(a)$.

Expected Value

- Expected value (mean): The probability-weighted average of the possible values,

$$\mu = E(X) = \sum_x x\, p(x).$$

- Multiply each value by its probability, then add everything up.
- Interpretations:
  - A weighted average: each value is weighted by how likely it is.
  - A measure of central tendency: a typical value of X.
  - A long-run average: the average over many independent repetitions.

Expected Value: Examples

- Daily sales X (in units) have the following PMF:

x	10	20	30
$p(x)$	0.3	0.5	0.2

$$E(X) = 10(0.3) + 20(0.5) + 30(0.2) = 3 + 10 + 6 = 19 \text{ units.}$$

  Note that 19 is not a possible value of X. The mean does not have to be in the support.
- Roulette, bet on red: $E(r) = 0 \times 19/37 + 1 \times 18/37 = 18/37 \approx 0.486$.
- Roulette, number: $E(b) = (0 + 1 + \cdots + 36)/37 = 18$.

The House Advantage

→ Expected value

X = Revenue from Crop harvest

	High P. $4	Low P. $1
Good H. 100kg	0.1	0.3
Bad H. 50kg	0.4	0.2

1) support $\{50, 100, 200, 400\}$

2) $E(X) = ?$ $E(X) = 10 + 30 + 80 + 40 = 160$

3) $E(X \mid \text{Good H.}) = ?$

- Bet \$1 on 14:

$$E(y_{14}) = (-1) \times \frac{36}{37} + 35 \times \frac{1}{37} = -\frac{1}{37} \approx -0.027.$$

$P(H,P \mid G,4) = \frac{P(H,P \cap G,H)}{P(G,H)} = \frac{0.1}{0.4}$

$P(L,P \mid G,H) = \frac{0.3}{0.4} = \frac{3}{4}$

$E(X \mid G,H) = \$100 \cdot P(L,P \mid G,H) + \$400 \cdot P(H,P \mid G,H)$

$95 + 100$

- Bet \$1 on red:

$$E(y_{Red}) = (-1) \times \frac{19}{37} + 1 \times \frac{18}{37} = -\frac{1}{37} \approx -0.027.$$

- Both bets lose about 2.7 cents per dollar on average. This is the house advantage: over many bets, the casino earns about 2.7% of all money wagered.
- The two bets have the same mean, but they are clearly not the same bet. We need a measure of spread (next section).

$E(X \mid \text{Good H.})$

⇒ Conditional Expected value of the return given that the harvest is 'Good'

Functions of a Random Variable

$Y = X + 1$ ← Random Variable

Now let's define a new random variable Y, Revenue of farming 100 acres $\Rightarrow Y = 100 \times X + 20$

$E(Y) = (100 \times \$50 \times P(X=50) + (100 \times \$100 \times P(X=100) + \cdots$ (+20) (+20)

- Any function of a random variable is also a random variable. If X is random, so are X^2, $\ln X$, and $Y = a + bX$.
- Expected value of a function: Use the PMF of X,

$$E[g(X)] = \sum_x g(x)\, p(x).$$

- Example: $E(y_{14}^2) = (-1)^2 \times \frac{36}{37} + 35^2 \times \frac{1}{37} = \frac{1261}{37} \approx 34.08$.
- Linear function: $Y = a + bX$ for constants a and b. Otherwise, it is a nonlinear function.
- Economic models are full of functions of random variables: revenue, profit, and utility all depend on random quantities.

Properties of Expected Values

$E(Y) = 100 \cdot \$50 \cdot P(X=50) + 100 \cdot \$100 \cdot P(X=100) + \cdots$

$= 100\,(\$50 \cdot P(X=50) + \$100 \cdot P(X=100) + \cdots)$

Suppose

$P = 3 \cdot X + 4 \cdot Y + 100$

$= 3X + 4Y + 100$

X = profit from growing corn ($/acre)

Y = profit from growing soybean ($/acre)

$E(X) = \$200/\text{acre}$

$E(Y) = \$150/\text{acre}$

If a farm grows 3 acres of corn, 4 acres of bean, expected profit, this from Government

For constants a, b and random variables X, Y:

- $E(a) = a$.
- Linearity: $E(a + bX) = a + b\,E(X)$. We only need $E(X)$, not the whole distribution.
- $E(X + Y) = E(X) + E(Y)$. This holds whether or not X and Y are independent.
- Roulette: $y_{Red} = 2r - 1$, so $E(y_{Red}) = 2E(r) - 1 = 2(18/37) - 1 = -1/37$.
- Never take the expected value inside a nonlinear function: in general $E[g(X)] \neq g(E(X))$.
  - $E(X^2) \neq [E(X)]^2$, $\quad E(1/X) \neq 1/E(X)$, $\quad E(\ln X) \neq \ln E(X)$.
  - Roulette: $E(b) = 18$ is red, but that does not mean a bet on red is expected to win.

$E(P) = 3E(X) + 4E(Y) + 100$

$= 600 + 600 + 100$

$= 1300$

Expected Value and Decisions

$E(a) = 500 \times 1 = 500$ $E(b) = 0 \cdot 0.5 + 1000 \cdot 0.5 = 500$

- Choose: (a) 500 for sure, or (b) a coin flip for 0 or 1,000?
- Both have the same expected value, but most people choose (a). They are risk-averse.
- Expected utility: People weigh outcomes by a utility function $u(x)$,

$$E[u(X)] = \sum_x u(x)\, p(x).$$

  A concave u penalizes spread. Since u is nonlinear, $E[u(X)] \neq u(E(X))$.
- St. Petersburg game: Win 2^k if the first head appears on flip k. The expected value is infinite, yet few people would pay much to play. Bernoulli's resolution (1738) founded expected utility theory.

Variance and Standard Deviation

- Variance: The expected squared deviation from the mean,

$$\sigma^2 = \mathrm{Var}(X) = E[(X-\mu)^2] = \sum_x (x-\mu)^2\, p(x).$$

- Shortcut formula (often easier):

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \sum_x x^2\, p(x) - \mu^2.$$

- Standard deviation: $\sigma = \sqrt{\mathrm{Var}(X)}$. It is measured in the same units as X, which makes it easier to interpret.
- The variance is always non-negative. Low variance means similar results each time; high variance means results vary a lot. In economics and finance, the variance of returns is the standard measure of risk.

$$\sigma^2 = E\left[(X-\mu)^2\right] \rightarrow \sum_x (x-\mu)^2 \cdot p(x)$$

($E(x)$)

Squared deviation from mean

$E(g(x))$ $\llcorner (x-\mu)^2$		HP	LP
	GH	0.1	0.4
	BH	0.3	0.2

Suppose = {\$50, \$100, \$150, \$200}

$E(x) = \$50 \cdot 0.2 + \$100 \cdot 0.4 + \cdots + \$400 \cdot 0.1$

$= \$110$

$Var(x) = E(X-110)^2$

$(\$50 - 150)^2 \cdot P(X=50)$

$+ (\$100 - 150)^2 \cdot P(X=100) \cdots$

$+ (\$200 - 150)^2 \cdot P(X=150)$

$+ (\$400 - 150)^2 \cdot P(X=200)$

$= \$100^2 \cdot 0.2 + \$60^2 \cdot 0.4 + \$50^2 \cdot 0.3 + \$250^2 \cdot 0.1$

$= 10,000$

$\sigma = \$100$

$Var(X) = E(X^2) - [E(X)]^2$

$= \$32,500 - (\$150)^2$

$22,500$

$= \$10,000$

$E(X) = \$150$

$E(X^2) = \$50^2 \cdot 0.2 = 500$

$+ \$100^2 \cdot 0.4 = 4000$

$+ \$200^2 \cdot 0.3 = 12000$

$+ \$400^2 \cdot 0.1 = 16000$

$= 32500$

Variance: An Example

A coffee shop sells X specialty drinks per hour, with the following PMF.

x	$p(x)$	$x\,p(x)$	$(x-\mu)^2\,p(x)$	$x^2\,p(x)$
10	0.1	1	$100 \times 0.1 = 10$	10
15	0.2	3	$25 \times 0.2 = 5$	45
20	0.4	8	$0 \times 0.4 = 0$	160
25	0.2	5	$25 \times 0.2 = 5$	125
30	0.1	3	$100 \times 0.1 = 10$	90
Sum	1	$\mu = 20$	$\sigma^2 = 30$	430

- Definition: $\sigma^2 = 10 + 5 + 0 + 5 + 10 = 30$.
- Shortcut: $\sigma^2 = 430 - 20^2 = 30$. Either way, $\sigma = \sqrt{30} \approx 5.48$ drinks.

Red or 14? Same Mean, Different Risk

- r is 0 or 1: $\mathrm{Var}(r) = (0 - 0.486)^2(0.514) + (1 - 0.486)^2(0.486) \approx 0.25$.
- y_{14}: Using the shortcut and $E(y_{14}^2) \approx 34.08$,

$$\mathrm{Var}(y_{14}) = 34.08 - (-0.027)^2 \approx 34.08, \qquad \sigma_{14} \approx 5.84.$$

- $y_{Red} = 2r - 1$: $\mathrm{Var}(y_{Red}) = 4 \times 0.25 \approx 1.0$, $\sigma_{Red} \approx 1.0$ (rule on the next slide).
- Both bets lose 2.7 cents per dollar on average, but:
  - Players who bet on red have similar results: most lose their money slowly.
  - Players who bet on 14 have very different results: a few win big, most lose quickly.

Variance of a Linear Function

$\mathrm{Var}(a+bX) = b^2\mathrm{Var}(X)$

Y

$\mathrm{Var}(Y) = E(Y-E(Y))^2$

$= E((a+bX) - (a+b(E(X))$

$= E((bX - bE(X))^2) = E[(b(X-E(X))^2] = E[b^2 \cdot (X-E(X))^2]$

- For constants a and b:

$$\mathrm{Var}(a+bX) = b^2\,\mathrm{Var}(X), \qquad \mathrm{sd}(a+bX) = |b|\,\mathrm{sd}(X).$$

- Adding a constant shifts the distribution but does not change its spread. Multiplying by b scales the variance by b^2. Also, $\mathrm{Var}(a) = 0$.
- Daily sales from before: $E(X) = 19$, $\mathrm{Var}(X) = 49$, $\sigma = 7$. Suppose each unit earns 5 and the fixed cost is 40, so profit is $Y = 5X - 40$.

$$E(Y) = 5(19) - 40 = 55, \qquad \mathrm{Var}(Y) = 5^2 \times 49 = 1225, \qquad \sigma_Y = 35.$$

- The fixed cost shifts expected profit but does not change risk.

Standardization

- Standardizing: Subtract the mean and divide by the standard deviation,

$$Z = \frac{X - \mu}{\sigma}.$$

- Z is a linear function of X, so by the rules above,

$$E(Z) = 0, \qquad \mathrm{Var}(Z) = \mathrm{sd}(Z) = 1.$$

- Z measures how many standard deviations X is from its mean. $Z = 0.5$: half a standard deviation above the mean. $Z = -3$: three standard deviations below.
- Daily sales: a day with 30 units has $Z = (30 - 19)/7 \approx 1.57$.
- We will use standardization heavily with the normal distribution.

Insurance: An Example

An insurer covers a loss of 10,000 that occurs with probability 0.02. Let X be the payout.

- $E(X) = 10000 \times 0.02 = 200$. This is the actuarially fair premium.
- $E(X^2) = 10000^2 \times 0.02 = 2{,}000{,}000$, so

$$\mathrm{Var}(X) = 2{,}000{,}000 - 200^2 = 1{,}960{,}000, \qquad \sigma = 1400.$$

- The standard deviation is seven times the mean: a policy is a very risky asset for the insurer to hold on its own.
- The premium actually charged exceeds 200 to pay for risk-bearing and administration. Customers pay it anyway because they are risk-averse.

Standard Distributions

- Some distributions appear so often in applications that they have names. We write

$$X \sim \text{DistributionName(parameters)},$$

  where $\sim$ reads "has the distribution" and the parameters are numbers that pin down the exact PMF.
- Each named distribution is a family: a different PMF for each value of the parameters.
- We derive the mean and variance once, and then use the results every time we recognize the distribution.
- Today: Bernoulli, binomial, Poisson, and discrete uniform.

Bernoulli Distribution

- $X \sim \text{Bernoulli}(p)$: $X = 1$ (success) with probability p and $X = 0$ (failure) with probability $1 - p$.
- Mean and variance:

$$E(X) = 0(1-p) + 1(p) = p, \qquad \text{Var}(X) = E(X^2) - p^2 = p - p^2 = p(1-p).$$

- Any indicator of an event A is Bernoulli with $p = P(A)$. Roulette: $r \sim \text{Bernoulli}(18/37)$.
- The workhorse for binary outcomes: employed or not, default or not, buy or not.
- The variance $p(1-p)$ is largest at $p = 0.5$, where uncertainty is greatest.

Binomial Distribution

- A binomial experiment has four features:
  1. A fixed number of trials, n.
  2. Each trial has two outcomes, success or failure.
  3. The probability of success, p, is the same on every trial.
  4. The trials are independent.
- $X \sim \mathrm{Binomial}(n, p)$: the number of successes in the n trials,

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n, \qquad \binom{n}{x} = \frac{n!}{x!(n-x)!}.$$

- Why: each sequence with x successes has probability $p^x(1-p)^{n-x}$ by independence, and there are $\binom{n}{x}$ such sequences.
- Mean and variance: $E(X) = np, \quad \mathrm{Var}(X) = np(1-p)$.

Binomial: Quality Control

A factory produces light bulbs with a 5% defect rate. In a sample of 20 bulbs, let $X =$ the number of defective bulbs, so $X \sim \text{Binomial}(20, 0.05)$.

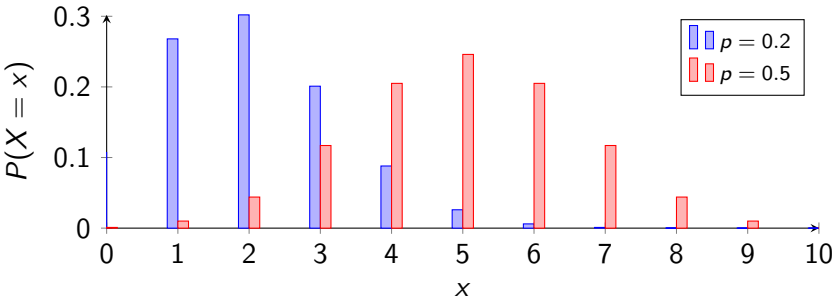
- Exactly 2 defective:

$$P(X = 2) = \binom{20}{2}(0.05)^2(0.95)^{18} = 190 \times 0.0025 \times 0.3972 \approx 0.189.$$

- Expected number of defectives: $E(X) = 20 \times 0.05 = 1$.
- Standard deviation: $\sigma = \sqrt{20 \times 0.05 \times 0.95} \approx 0.975$.
- At least one defective: use the complement rule,

$$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.95^{20} \approx 0.64.$$

Binomial Shapes



- With $n = 10$, the binomial is right-skewed for $p = 0.2$ and symmetric for $p = 0.5$.
- As n grows, every binomial becomes bell-shaped. This previews the normal approximation and the Central Limit Theorem.

When the Binomial Fails

- The binomial requires independent trials with the same p. Check both before using it.
- Sampling without replacement from a small population changes p from draw to draw (recall the M&Ms in Lecture 4). The hypergeometric distribution handles this case.
- Loan portfolio: 100 loans, each defaulting with $p = 0.05$. If defaults were independent, $\sigma = \sqrt{100 \times 0.05 \times 0.95} \approx 2.2$, and 20 or more defaults would be essentially impossible.
- But defaults share common shocks such as recessions, so they are not independent. The number of defaults is then far more variable, and 20 or more becomes plausible.
- Assuming independence when it fails understates risk. This was a central lesson of the 2008 financial crisis.

Poisson Distribution

- $X \sim \text{Poisson}(\lambda)$: the number of events in a fixed interval of time or space, when events occur independently at a constant average rate λ,

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

- Mean and variance: $E(X) = \text{Var}(X) = \lambda$. The mean equals the variance.
- Examples: customers arriving per hour, insurance claims per day, traffic accidents per week, defects per square meter.
- The rate must match the interval. If there are 2 accidents per week on average, the number in two weeks is Poisson with $\lambda = 4$.
- The Poisson approximates the binomial when n is large and p is small, with $\lambda = np$.

Poisson: Examples

- A bank sees an average of 3 customers every 15 minutes at lunch. $X \sim \text{Poisson}(3)$.

$$P(X = 2) = \frac{e^{-3}3^2}{2!} \approx 0.224, \qquad P(X \leq 2) = e^{-3}\left(1 + 3 + \frac{9}{2}\right) \approx 0.423.$$

  The standard deviation is $\sqrt{3} \approx 1.73$ customers.
- A call center receives an average of 4 complaints per day. $X \sim \text{Poisson}(4)$.
  - $P(\text{no complaints}) = e^{-4} \approx 0.018$.
  - Staffing: $P(X > 8) = 1 - P(X \leq 8) \approx 0.021$. A team sized for 8 complaints is overwhelmed on about 2% of days.
- Excel: `POISSON.DIST(x,λ,FALSE)`. R: `dpois(x,lambda)` and `ppois(x,lambda)`.

Discrete Uniform Distribution

- Discrete uniform: Puts equal probability on every value in a finite set S,

$$p(x) = \frac{1}{|S|} \quad \text{for } x \in S.$$

- This is the classical probability from Lecture 4: all outcomes equally likely.
- Roulette: b is discrete uniform on $\{0, 1, \ldots, 36\}$, with $p(x) = 1/37$ and $E(b) = 18$.
- A fair die: X is discrete uniform on $\{1, \ldots, 6\}$, with $E(X) = 3.5$ and

$$\mathrm{Var}(X) = \frac{1 + 4 + 9 + 16 + 25 + 36}{6} - 3.5^2 = \frac{91}{6} - 12.25 \approx 2.92.$$

Which Distribution?

Situation	Distribution	$E(X)$	$\mathrm{Var}(X)$
One binary outcome	Bernoulli(p)	p	$p(1-p)$
Successes in n independent trials, same p	Binomial(n, p)	np	$np(1-p)$
Events in an interval, constant rate	Poisson(λ)	λ	λ
Equally likely values in a finite set	Discrete uniform		

- Identify the experiment first. The distribution follows from its structure, not from habit.
- In R, the prefix `d` gives the PMF, `p` gives the CDF, and `r` gives random draws: e.g., `dbinom`, `pbinom`, `rbinom`; `dpois`, `ppois`, `rpois`.

Summary

- A random variable assigns a number to each random outcome. The PMF $p(x) = P(X = x)$ and the CDF $F(x) = P(X \leq x)$ each describe its distribution completely.
- Expected value: $E(X) = \sum_x x\, p(x)$. Linearity: $E(a + bX) = a + b\, E(X)$, but $E[g(X)] \neq g(E(X))$ for nonlinear g.
- Variance: $\mathrm{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$, and $\mathrm{Var}(a + bX) = b^2\, \mathrm{Var}(X)$.
- Two random variables can have the same mean but very different risk.
- Bernoulli for one binary outcome, binomial for the number of successes in n independent trials, Poisson for counts of events at a constant rate, discrete uniform for equally likely values.

Practice Question 1: A Vendor's Profit

A street vendor's daily profit X has the following PMF.

x	-10	20	50
$p(x)$	0.2	0.5	0.3

(a) Verify that this is a valid PMF, and find the CDF $F(x)$ at $x = -10, 20, 50$.

(b) What is the probability that the vendor makes a positive profit?

(c) Find $E(X)$.

(d) Find $\mathrm{Var}(X)$ and the standard deviation of X.

(e) The vendor pays a 20% tax on profit and a fixed stall fee of 5 per day, so the vendor's take-home pay is $Y = 0.8X - 5$. Find $E(Y)$ and the standard deviation of Y.

Practice Question 1: Answers

(a) Each $p(x)$ is between 0 and 1, and $0.2 + 0.5 + 0.3 = 1$. $F(-10) = 0.2$, $F(20) = 0.7$, $F(50) = 1$.

(b) $P(X > 0) = 1 - F(-10) = 0.8$.

(c) $E(X) = -10(0.2) + 20(0.5) + 50(0.3) = -2 + 10 + 15 = 23$.

(d) $E(X^2) = 100(0.2) + 400(0.5) + 2500(0.3) = 970$, so

$$\mathrm{Var}(X) = 970 - 23^2 = 441, \qquad \sigma = 21.$$

(e) $E(Y) = 0.8(23) - 5 = 13.4$, $\quad \mathrm{sd}(Y) = 0.8 \times 21 = 16.8$.

Practice Question 2: Target Practice

A marksman hits a target 80% of the time. The marksman fires 6 shots, and the shots are independent. Let X = the number of shots that hit the target.

(a) What is the distribution of X? State its parameters.

(b) What is the probability that exactly 4 shots hit the target?

(c) What is the probability that more than 4 shots hit the target?

(d) What is the probability that at least 2 shots hit the target?

(e) Find the mean and standard deviation of X. Would it be unusual for none of the shots to hit the target?

Practice Question 2: Answers

(a) $X \sim \text{Binomial}(6, 0.8)$.

(b) $P(X = 4) = \binom{6}{4}(0.8)^4(0.2)^2 = 15 \times 0.4096 \times 0.04 \approx 0.246$.

(c) $P(X > 4) = P(X = 5) + P(X = 6) = 0.393 + 0.262 \approx 0.655$.

(d) Complement rule:

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0.000064 - 0.001536 = 0.9984.$$

(e) $E(X) = 6 \times 0.8 = 4.8$ and $\sigma = \sqrt{6 \times 0.8 \times 0.2} \approx 0.98$. Zero hits is $(0 - 4.8)/0.98 \approx -4.9$ standard deviations below the mean, and $P(X = 0) = 0.2^6 = 0.000064$. Yes, it would be very unusual.

Practice Question 3: Traffic Accidents

A section of highway has on average 2 traffic accidents per week. Assume that accidents occur independently at a constant rate.

(a) What is the distribution of the number of accidents in one week?

(b) What is the probability of exactly one accident in a week?

(c) What is the probability of no accidents in a week?

(d) What is the probability of at most three accidents in a two-week period?

Practice Question 3: Answers

(a) $X \sim \text{Poisson}(2)$, so $E(X) = \text{Var}(X) = 2$.

(b) $P(X = 1) = e^{-2}\, 2^1/1! = 2e^{-2} \approx 0.271$.

(c) $P(X = 0) = e^{-2} \approx 0.135$.

(d) Over two weeks, the rate is $\lambda = 2 \times 2 = 4$, so $Y \sim \text{Poisson}(4)$:

$$P(Y \leq 3) = e^{-4}\left(1 + 4 + \frac{4^2}{2!} + \frac{4^3}{3!}\right) = e^{-4}(1 + 4 + 8 + 10.67) \approx 0.433.$$